



**Verified Carbon  
Standard**

## D.LIGHT'S IMPROVED COOKING PROJECT IN KENYA



Document Prepared by



**Climate Secure**

|                      |                                                                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b> | D.LIGHT'S IMPROVED COOKING PROJECT IN KENYA                                                                                                                                                                                                           |
| <b>Version</b>       | 06.0                                                                                                                                                                                                                                                  |
| <b>Date of Issue</b> | 14-Mar-2024                                                                                                                                                                                                                                           |
| <b>Prepared By</b>   | Climate Secure                                                                                                                                                                                                                                        |
| <b>Contact</b>       | Address: P.O. Box 515000, Sharjah Media City, Sharjah, UAE<br>Telephone: +91 9717093552<br>Email: <a href="mailto:info@climate-secure.com">info@climate-secure.com</a><br>Website: <a href="http://www.climate-secure.com">www.climate-secure.com</a> |

# CONTENTS

---

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>PROJECT DETAILS .....</b>                                          | <b>4</b>  |
| 1.1      | Summary Description of the Project .....                              | 4         |
| 1.2      | Sectoral Scope and Project Type.....                                  | 4         |
| 1.3      | Project Eligibility .....                                             | 5         |
| 1.4      | Project Design .....                                                  | 6         |
| 1.5      | Project Proponent .....                                               | 7         |
| 1.6      | Other Entities Involved in the Project .....                          | 7         |
| 1.7      | Ownership .....                                                       | 7         |
| 1.8      | Project Start Date .....                                              | 7         |
| 1.9      | Project Crediting Period .....                                        | 7         |
| 1.10     | Project Scale and Estimated GHG Emission Reductions or Removals ..... | 8         |
| 1.11     | Description of the Project Activity.....                              | 8         |
| 1.12     | Project Location .....                                                | 11        |
| 1.13     | Conditions Prior to Project Initiation .....                          | 12        |
| 1.14     | Compliance with Laws, Statutes and Other Regulatory Frameworks .....  | 12        |
| 1.15     | Participation under Other GHG Programs .....                          | 12        |
| 1.16     | Other Forms of Credit .....                                           | 12        |
| 1.17     | Sustainable Development Contributions .....                           | 13        |
| 1.18     | Additional Information Relevant to the Project.....                   | 15        |
| <b>2</b> | <b>SAFEGUARDS .....</b>                                               | <b>15</b> |
| 2.1      | No Net Harm .....                                                     | 15        |
| 2.2      | Local Stakeholder Consultation .....                                  | 24        |
| 2.3      | Environmental Impact .....                                            | 26        |
| 2.4      | Public Comments .....                                                 | 26        |
| 2.5      | AFOLU-Specific Safeguards .....                                       | 26        |
| <b>3</b> | <b>APPLICATION OF METHODOLOGY .....</b>                               | <b>26</b> |
| 3.1      | Title and Reference of Methodology .....                              | 27        |
| 3.2      | Applicability of Methodology .....                                    | 27        |
| 3.3      | Project Boundary .....                                                | 30        |
| 3.4      | Baseline Scenario .....                                               | 31        |
| 3.5      | Additionality .....                                                   | 31        |

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| 3.6      | Methodology Deviations .....                                               | 34        |
| <b>4</b> | <b>QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS.....</b>         | <b>34</b> |
| 4.1      | Baseline Emissions .....                                                   | 34        |
| 4.2      | Project Emissions.....                                                     | 34        |
| 4.3      | Leakage.....                                                               | 34        |
| 4.4      | Net GHG Emission Reductions and Removals.....                              | 35        |
| <b>5</b> | <b>MONITORING .....</b>                                                    | <b>36</b> |
| 5.1      | Data and Parameters Available at Validation .....                          | 36        |
| 5.2      | Data and Parameters Monitored .....                                        | 39        |
| 5.3      | Monitoring Plan .....                                                      | 41        |
|          | Appendix 1: Non-Binding Mechanisms for Success of the Carbon project ..... | 45        |
|          | Appendix 2: fNRB Calculation as per CDM Tool 30, V4.0.....                 | 48        |

# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

The “D.LIGHT’S IMPROVED COOKING PROJECT IN KENYA” is a large-scale project that aims dissemination of energy-efficient improved cookstoves (ICS) to replace existing traditional Cookstoves in domestic households and communities in Kenya.

Traditional cooking methods prevalent across host country, contribute to greenhouse gas (GHG) emissions attributed to inefficient combustion of unsustainably sourced, non-renewable biomass (NRB) fuel. Furthermore, the use of solid biomass fuels (e.g., wood) in inefficient traditional stoves and/or open fires releases large amounts of particulate matter (PM), creating hazardous levels of indoor air pollution (IAP).

### Technologies/Measures

The project activity includes dissemination of woody biomass ICS to project households using traditional modes of cooking in the baseline. The ICSs disseminated under the project are designed to improve fuel combustion and heat transfer. The ICS results in the reduction of fuel consumption and improvement in levels of Indoor air pollution for project users (reducing smoke, black soot, and particulate matter (PM) emissions). The reduction in consumption of cooking fuel also reduces equivalent greenhouse gas (GHG) emissions attributed to the use of non-renewable biomass. The project activity intends to access carbon finance to mitigate the main barrier (price) for end users towards adopting clean cookstoves.

Expected Year wise distribution / installation of the project ICS is given below<sup>1</sup>:

| Year   | Proposed Implementation Plan (# ICS) |
|--------|--------------------------------------|
| Year 1 | 80,000                               |
| Year 2 | 120,000                              |
| Total  | 200,000                              |

In the absence of this project, the project beneficiaries would cook primarily using traditional inefficient stoves for meeting their thermal energy needs (the scenario existing prior to the project activity), perpetuating environmental and health degradation.

The average annual GHG emission reduction from the project activity is estimated to be 791,210 tCO<sub>2</sub>e and a total 5,538,473 tCO<sub>2</sub>e over the first crediting period of 7 years.

## 1.2 Sectoral Scope and Project Type

Mandatory Sectoral Scope: 03 – Energy Demand

Conditional Sectoral Scope: Not Applicable

Project Type: Energy Efficiency Project

<sup>1</sup> This is tentative/ / expected plan and may differ from actual implementation which depends upon market demand and user preferences, deemed outside the control of PP. The number of ICS units mentioned in the table is therefore only an indicative number for ex-ante calculations and is non-binding. The number of units implemented may change ex-post, based on types of stoves distributed and user adoption/preference. During the crediting period, more/lesser systems may get distributed owing to customer demand and shall not be deemed a deviation from the project description.

The project is not a grouped project, but a large-scale project with multiple project activity instances<sup>2</sup>.

### 1.3 Project Eligibility

**1. The Seven Kyoto Protocol GHGs:** The project is applicable to this scope, as the project involves emission reduction comes from Kyoto GHGs viz. CO<sub>2</sub>, CH<sub>4</sub> and N<sub>2</sub>O in the baseline.

**2. Ozone-depleting substances (ODS):** NA. The project does not involve ODSs.

**3. Project activities supported by a methodology approved under the VCS program through the methodology development and review process:** The project uses the methodology VMR006 version 1.1 which is approved under the VCS program through the methodology development and review process.

**4. Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded:** NA. The project does not use a methodology approved under other approved GHG program.

**5. Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements:** NA. The project does not involve REDD+.

Additionally, the project is not listed in table 1, section 2.1.3 of the VCS standard version 4.4, given it involves installation of improved cooking stoves (ICS) in Kenya.

The proposed project activity envisages distribution of 200,000 ICS in Kenya and meets compliance with section 2.1.3 of VCS Standard v (4.4).

Compliance with other general requirements is described below:

| Reference to VCS Standard version 4.4 | Requirement                                                                                                                                                                                                                                                                      | Justification                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1.1                                 | Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.                                                                                           | The project meets all applicable rules and requirements as set out under the VCS Program and is guided by the ISO 14064-2:2006 clause 3.                          |
| 3.1.2                                 | Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1.                                  | The project applies VCS approved methodology VMR0006, version 1.1 along with tools or modules as applicable. Refer to section 3 of the PD below for more details. |
| 3.1.3                                 | Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.21. Projects must update to the latest version of the methodology when reassessing the baseline and renewing a crediting period. | The project applies the latest version 1.1 of VCS approved methodology VMR0006. Refer to section 3 of the PD below for more details.                              |

<sup>2</sup> Each ICS unit is deemed as a project activity instance.

|       |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                          |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1.4 | Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.                                                                                                                                                                                                                                                               | The project and its instances shall remain in compliance with the applicable local laws. This has been made an eligibility criterion for inclusion of project activity into the project. |
| 3.1.5 | Where projects apply methodologies that permit the project proponent its own choice of model, such model shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or removals). | Not applicable. The applied methodology VMR0006 version 1.1 does not provide any such option.                                                                                            |
| 3.1.6 | Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements.                                       | Not applicable. The applied methodology VMR0006 version 1.1 does not provide any such choice.                                                                                            |
| 3.1.7 | Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.                                                                                                                                                                                                                             | The project activity applies VCS methodology and so this is not applicable.                                                                                                              |
| 3.1.8 | Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.                                                                                                                                         | The project activity applies VCS methodology i.e., VMR0006 version 1.1 and so this is not applicable.                                                                                    |
| 3.1.9 | Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 of VCS standard ver 4.4 and Effective Dates or equivalent for other program documents, and are listed in a companion Summary of Effective Dates document which corresponds with each update                                                                                                                    | The project will adhere to the new VCS Program rules, as applicable and set out in Appendix 3 of VCS standard.                                                                           |

## 1.4 Project Design

The proposed large scale project activity involves multiple project activity instances (PAI) in the geographical boundary of Kenya but is not being developed as a grouped project. All PAI (project ICS) shall remain limited to the geographical boundary of Kenya.

### Eligibility Criteria

The project is not a grouped project; hence this is not applicable.

## 1.5 Project Proponent

|                          |                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------|
| <b>Organization name</b> | d.light design, ltd                                                               |
| <b>Contact person</b>    | Karl Skare                                                                        |
| <b>Title</b>             | Managing Director - Global Partnerships & Strategy                                |
| <b>Address</b>           | 2100 Geng Road, Suite 210, Palo Alto, California, 94303, United States of America |
| <b>Telephone</b>         | +1 513-543-9849                                                                   |
| <b>Email</b>             | Karl.skare@dlight.com                                                             |

## 1.6 Other Entities Involved in the Project

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>Organization name</b>   | Climate Secure                                     |
| <b>Role in the project</b> | Entity Completing the Project Description Document |
| <b>Contact person</b>      | Rohit Lohia                                        |
| <b>Title</b>               | Director                                           |
| <b>Address</b>             | P.O. Box 515000, Sharjah Media City, Sharjah, UAE  |
| <b>Telephone</b>           | +91 9717093552                                     |
| <b>Email</b>               | rohit.lohia@climate-secure.com                     |

## 1.7 Ownership

For each PAI, the ownership of the instance shall be clearly demonstrated by virtue of the following:

- Declaration / agreement with end user transferring the ownership rights of the carbon credits generated from the project activity instance (ICS) to the project proponent.

Thus, all VCUs generated from the project activity instances shall be owned by d.light design, ltd hereinafter referred as "d.light", unless otherwise stated.

## 1.8 Project Start Date

08 September 2022 (Date for installation of 1<sup>st</sup> ICS under the project).

## 1.9 Project Crediting Period

First Crediting Period start date: 08 September 2022

First Crediting Period end date: 07 September 2029

Crediting Period Duration: 21 years (7 Years, Renewable twice)

The date of first ICS distribution under the project activity shall mark the start date of crediting period for the project.

## 1.10 Project Scale and Estimated GHG Emission Reductions or Removals

| Project Scale |   |
|---------------|---|
| Project       |   |
| Large project | X |

| Year                                   | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|----------------------------------------|--------------------------------------------------------------------|
| Year 1                                 | 365,221                                                            |
| Year 2                                 | 898,282                                                            |
| Year 3                                 | 883,659                                                            |
| Year 4                                 | 869,183                                                            |
| Year 5                                 | 854,851                                                            |
| Year 6                                 | 840,662                                                            |
| Year 7                                 | 826,615                                                            |
| <b>Total estimated ERs</b>             | <b>5,538,473</b>                                                   |
| <b>Total number of crediting years</b> | <b>7</b>                                                           |
| <b>Average annual ERs</b>              | <b>791,210</b>                                                     |

## 1.11 Description of the Project Activity

The purpose of the project activity is to disseminate energy-efficient, improved cookstoves (ICS) to replace existing traditional cookstoves in domestic households and communities in Kenya.

In Kenya, 63.3% of households use some type of solid fuel for cooking, with 51.6% and 11.4% of households utilizing firewood and charcoal respectively, as primary fuel. Fuelwood is prevalent in rural areas of Kenya with 80.1% rural households dependent on firewood for cooking, while only 7.7% rural household dependent on charcoal for cooking. On the other hand, urban areas of Kenya have relatively charcoal prevalence with 16.9 % of urban households using charcoal while only 9.2% using fuelwood for cooking<sup>3</sup>.

With respect to cooking technology, 51.2% of Kenya's households use traditional/inefficient stoves for cooking (three stone stove/open fire and/or traditional solid fuel stove without chimney). 79.5% of rural and 9.3% of urban households use traditional/inefficient stoves as cooking technology. This is evidenced by the latest available Demographic and Health Survey Report (DHS) 2022, published in June 2023 by the Kenya National Bureau of Statistics (KNBS) in collaboration with the Ministry of Health (MoH), supported by funding by the Government of Kenya and various multilateral organizations.

<sup>3</sup> Table 2.2, page 27, DHS 2022, published in June 2023, <https://dhsprogram.com/pubs/pdf/FR380/FR380bis.pdf>



**Table 2.2 Household characteristics: Cooking**

Percent distribution of households and de jure population by place for cooking, cooking technology, and cooking fuel, according to residence, Kenya DHS 2022

| Characteristic                                  | Households    |               |               | Population    |               |                |
|-------------------------------------------------|---------------|---------------|---------------|---------------|---------------|----------------|
|                                                 | Urban         | Rural         | Total         | Urban         | Rural         | Total          |
| <b>Place for cooking</b>                        |               |               |               |               |               |                |
| In the house                                    | 84.9          | 26.0          | 49.7          | 80.0          | 20.4          | 40.5           |
| Separate room/kitchen                           | 31.9          | 11.4          | 19.7          | 38.0          | 10.0          | 19.4           |
| No separate room/kitchen                        | 52.9          | 14.6          | 30.1          | 42.0          | 10.4          | 21.0           |
| In a separate building                          | 7.7           | 63.1          | 40.8          | 12.4          | 69.4          | 50.3           |
| Outdoors                                        | 5.5           | 9.6           | 7.9           | 6.9           | 9.7           | 8.7            |
| No food cooked in household                     | 1.9           | 1.3           | 1.6           | 0.7           | 0.4           | 0.5            |
| <b>Total</b>                                    | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>   |
| <b>Main cooking technology</b>                  |               |               |               |               |               |                |
| <b>Clean fuels and technologies</b>             | <b>63.3</b>   | <b>9.6</b>    | <b>31.2</b>   | <b>58.7</b>   | <b>6.0</b>    | <b>23.7</b>    |
| Electric stove                                  | 0.4           | 0.1           | 0.2           | 0.3           | 0.1           | 0.2            |
| LPG/natural gas stove                           | 59.9          | 8.8           | 29.4          | 55.7          | 5.5           | 22.4           |
| Piped natural gas stove                         | 2.1           | 0.4           | 1.1           | 1.9           | 0.2           | 0.8            |
| Biogas stove                                    | 0.3           | 0.2           | 0.2           | 0.3           | 0.1           | 0.2            |
| Liquid fuel stove using alcohol/ethanol         | 0.6           | 0.0           | 0.2           | 0.5           | 0.0           | 0.2            |
| <b>Other fuels and technologies</b>             | <b>34.8</b>   | <b>89.1</b>   | <b>67.2</b>   | <b>40.6</b>   | <b>93.6</b>   | <b>75.8</b>    |
| Liquid fuel stove not using alcohol/ethanol     | 8.4           | 0.6           | 3.7           | 6.5           | 0.2           | 2.3            |
| Manufactured solid fuel stove                   | 16.9          | 7.4           | 11.3          | 20.2          | 6.2           | 10.9           |
| With a chimney                                  | 2.4           | 1.3           | 1.7           | 3.1           | 1.1           | 1.8            |
| Without a chimney                               | 14.5          | 6.2           | 9.5           | 17.1          | 5.1           | 9.1            |
| <b>Traditional solid fuel stove</b>             | <b>0.7</b>    | <b>4.7</b>    | <b>3.1</b>    | <b>0.8</b>    | <b>5.5</b>    | <b>3.9</b>     |
| With a chimney                                  | 0.2           | 1.5           | 1.0           | 0.2           | 1.8           | 1.3            |
| Without a chimney                               | 0.5           | 3.2           | 2.1           | 0.6           | 3.7           | 2.7            |
| <b>Three stone stove/open fire</b>              | <b>8.8</b>    | <b>76.3</b>   | <b>49.1</b>   | <b>13.1</b>   | <b>81.7</b>   | <b>58.6</b>    |
| <b>No food cooked in household</b>              | <b>1.9</b>    | <b>1.3</b>    | <b>1.6</b>    | <b>0.7</b>    | <b>0.4</b>    | <b>0.5</b>     |
| <b>Total</b>                                    | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>   |
| <b>Cooking fuel</b>                             |               |               |               |               |               |                |
| <b>Clean fuels and technologies<sup>1</sup></b> | <b>63.3</b>   | <b>9.6</b>    | <b>31.2</b>   | <b>58.7</b>   | <b>6.0</b>    | <b>23.7</b>    |
| <b>Solid fuels for cooking</b>                  | <b>26.2</b>   | <b>88.4</b>   | <b>63.3</b>   | <b>34.0</b>   | <b>93.3</b>   | <b>73.4</b>    |
| Charcoal                                        | 16.9          | 7.7           | 11.4          | 20.1          | 6.5           | 11.0           |
| Wood                                            | 9.2           | 80.1          | 51.6          | 13.8          | 86.3          | 61.9           |
| Straw/shrubs/grass                              | 0.1           | 0.3           | 0.2           | 0.1           | 0.3           | 0.2            |
| Agricultural crop                               | 0.0           | 0.1           | 0.1           | 0.0           | 0.1           | 0.1            |
| Sawdust                                         | 0.0           | 0.1           | 0.1           | 0.1           | 0.1           | 0.1            |
| <b>Other fuels</b>                              | <b>8.6</b>    | <b>0.7</b>    | <b>3.9</b>    | <b>6.6</b>    | <b>0.3</b>    | <b>2.4</b>     |
| Gasoline/diesel                                 | 0.1           | 0.0           | 0.0           | 0.1           | 0.0           | 0.0            |
| Kerosene/paraffin                               | 8.4           | 0.7           | 3.8           | 6.5           | 0.3           | 2.4            |
| <b>No food cooked in household</b>              | <b>1.9</b>    | <b>1.3</b>    | <b>1.6</b>    | <b>0.7</b>    | <b>0.4</b>    | <b>0.5</b>     |
| <b>Total</b>                                    | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>  | <b>100.0</b>   |
| <b>Number of households/population</b>          | <b>15,277</b> | <b>22,634</b> | <b>37,911</b> | <b>47,730</b> | <b>94,296</b> | <b>142,026</b> |

LPG = liquefied petroleum gas

<sup>1</sup> Includes stoves/cookers using electricity, LPG/natural gas/biogas, solar, and alcohol/ethanol

Thus, cooking using solid biomass fuel over traditional inefficient stoves is the prominent cooking practice.

The use of these traditional inefficient biomass stoves is a significant driver of deforestation. Widespread usage of unsustainably sourced, non-renewable biomass (NRB) for cooking, results in equivalent GHG emissions attributed to use of non-renewable biomass.

Additionally, the use of solid biomass fuels (e.g., wood) in inefficient traditional stoves and/or open fires releases large amounts of particulate matter (PM), creating hazardous levels of indoor air pollution (IAP). IAP generated by inefficient and poorly ventilated stoves burning biomass fuels is responsible for the deaths of an estimated 3.2 million people annually. Almost half of these deaths occur among children under five years of age. Exposure is high among women and children, who spend the most time near the domestic hearth. Reliance on polluting fuels and technologies also requires significant time for cooking on an inefficient device and gathering and preparing fuel. In developing countries with high mortality rates overall, indoor air pollution ranks fourth in terms of the risk factors that contribute to disease and death (World Health Organization)<sup>4</sup>.

<sup>4</sup> <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

### Technology/Measure

The ICS disseminated under the project is designed to improve fuel combustion and heat transfer. It results in the reduction of fuel consumption and improvement in levels of Indoor air pollution for project users (reducing smoke, black soot, and particulate matter (PM)). The reduction in consumption of cooking fuel also reduces greenhouse gas (GHG) emissions attributed to the use of non-renewable biomass.

The project includes dissemination of following woody biomass ICS<sup>5</sup>, besides others. Specification of these example ICS<sup>6</sup> is given below:

| Stove Model | Photo                                                                               | Manufacturer | Thermal Efficiency                               | Estimate Life | Fixed / portable | Grate / Chimney |
|-------------|-------------------------------------------------------------------------------------|--------------|--------------------------------------------------|---------------|------------------|-----------------|
| Kuniokoa    |    | Burn         | 51.3% with Pot-skirt;<br>43.4% without Pot-skirt | Upto 7 years  | Portable         | Grate           |
| Ecoa Wood   |    | Burn         | 51.3% with Pot-skirt;<br>43% without Pot-skirt   | Upto 7 years  | Portable         | Grate           |
| Jikokoa     |   | Burn         | 53.7%                                            | Upto 7 years  | Portable         | Grate           |
| Ecoa Char   |  | Burn         | 53.7%                                            | Upto 7 years  | Portable         | Grate           |

In the absence of this project activity, the beneficiaries shall cook primarily using traditional inefficient stoves / 3 stone fires, perpetuating environmental and health degradation.

The project intends to access carbon finance to mitigate the main barrier (price) for end users towards adopting clean cookstoves.

**Management Plan and Operational Framework:** The project will be implemented according to the following plan:

<sup>5</sup> The ICS listed are non-exhaustive. The ICS distributions are subject to potential customer demands and preferences. Other ICS models may be included in the Project subsequently, provided they meet the methodological requirement and technology/measure described in section 1.1, page 4 above. This shall not be construed as a deviation or change from project description.

<sup>6</sup> Project ICS may be distributed with or without pot-skirt depending upon PP's discretion in future. In case where pot-skirts are being distributed, the same shall be duly captured and recorded at the time of distribution. Accordingly, the corresponding thermal efficiency of the ICS with/without pot skirt shall be used for ER calculations. In such case, ex-post monitoring shall assess (on sampling basis) the continuity of usage of pot skirt and shall calculate ERs accordingly discounting for any drop off of pot skirts.

1. The PP is responsible for general management of the project, including appointing all personnel / groups of personnel involved in the project development and implementation.
2. The distribution of the ICS will be managed via PP's own distribution network or with local distribution partners.
3. The PP will communicate to the project beneficiaries during distribution that PP retains exclusive and unconditional rights to the carbon credit generated by use of project ICS.
4. The PP will maintain and update the ICS distribution database continuously. The required data collection includes, but is not limited to, the following.
  - a. Date of distribution
  - b. Geographic area of distribution
  - c. Model/type of project ICS distributed.
  - d. Quantity of project ICS distributed.
  - e. Name of the beneficiary, address, and telephone number (if available)
  - f. Unique serial number of project ICS
  - g. Primary Baseline fuel / technology being used.
5. The information collected will be stored in the distribution database will serve as the project database for ex-post project sampling and monitoring purposes. Also, this shall serve to avoid any double counting and duplicity of data.
6. The PP will coordinate ex-post monitoring activities as follows:
  - a. Implement the monitoring plan,
  - b. Determine the sample size as per sampling plan and identify the samples to be monitored (the sampling plan is detailed below)
  - c. Ex-post monitoring - the data will be collected using survey questionnaires designed by experts. The surveys will be carried out by external parties or by project implementer teams experience / trained to conduct such surveys. PP will be responsible for ensuring that staff responsible for monitoring and reporting have received adequate training, as applicable.
  - d. Ensure the quality of monitoring data (QA/QC) as per section 5.2 below. The monitoring results and statistical robustness will be reviewed internally and approved by competent staff.
  - e. Use the data collected for emissions reduction calculations.

The aforesaid plan shall ensure that the project is managed and operated in a standard manner.

## 1.12 Project Location

**Host Country:** Kenya

**Region/State/Province etc.:** All regions of Kenya

**City/Town/Community etc.:** All cities and towns in Kenya

**Physical/Geographical location:**

Kenya is spread from 5.03° N to -4.65° N latitude, while the longitude spread is between 34.03° E to 41.9° E. With the land area of 581,309 km<sup>2</sup>. The capital of Kenya is Nairobi, and its geographic coordinates are 1°16' S latitude and 36°48' E longitude. The map of Kenya, for reference purpose, is as follows:



**Figure 3 Map of Kenya**

### 1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable biomass in traditional/inefficient baseline stoves by the target population to meet equivalent thermal energy needs as provided by project cookstoves in absence of project activity. As specified in Section 1.11 above, the use of traditional means of cooking is prevalent in Kenya since long and hence project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

The baseline scenario is same as the condition existing prior to the project.

### 1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Kenya.

### 1.15 Participation under Other GHG Programs

#### 1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered, nor is it seeking registration under any other GHG program.

#### 1.15.2 Projects Rejected by Other GHG Programs

The project activity has not been rejected by any other GHG program.

### 1.16 Other Forms of Credit

#### 1.16.1 Emissions Trading Programs and Other Binding Limits

The project activity is a voluntary initiative, and it is not obligated to meet any local laws or regulatory compliances. The project does not reduce GHG emissions from activities, that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

### 1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related credit, including renewable energy certificates.

### Supply Chain (Scope 3) Emissions

This is not applicable, since project's GHG emissions reductions or removals do not occur within a supply chain but at the project beneficiary location. Also, the project beneficiary(ies) has assigned unconditional rights to the ownership of credits to d.light, precluding anyone other than d.light to claim concerned credits.

### 1.17 Sustainable Development Contributions

Kenya is one of the United Nations member states that has committed to achieve Sustainable Development Goals. The project contributes to social, environmental, and economic benefits which contribute to sustainable development of the local environment and the country.

The project contributions to the relevant SDGs are discussed below:

| SDG                            | Project Contributions - Justification                                                                                                                                                                                                                                                                                                                                                                                            | Relevant SDG Target                                                                                                                                                                                                                                                                                                                         | SDG Impact Indicator                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - No Poverty                 | <b>Economic Sustainability:</b> Household expenditures on purchased cooking fuel will get reduced with usage of project ICS owing to its increased efficiency. Thus, project ICS shall result in money savings in beneficiary households. The project will distribute ICS and will lead to reduction of poverty in the beneficiary households.                                                                                   | 1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance | <b>Project Specific Indicator / parameter:</b> Average monthly household savings due to decrease in expenditure on purchased fuel after shifting to project technology. |
| 3 - Good Health and Well Being | <b>Social Sustainability:</b> The indoor pollutants emissions attributed to inefficient burning of solid biomass fuels in the beneficiary households will be reduced.<br><br>The project ICS shall aid improved combustion of solid biomass fuel thereby reducing smoke and Particulate matter (PM) emissions. The project will free the beneficiary households from exposure to smoke / PM associated with traditional cooking. | 3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.                                                                                                                                                                                          | <b>Project Specific Indicator / parameter:</b> % users reporting reduction in smoke/PM after shifting to ICS in project.                                                |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 - Quality Education           | <p><b>Social Sustainability-</b> The project activity is targeting to provide formal and non-formal education and training to local people under the project activity.</p> <p>In this project activity, the local distribution team will be trained on improved cookstoves and the end-user HHs will trained about sustainable cooking practices.</p>                                                                                                                                                                                                                                                                                             | 4.3 By 2030, ensure equal access for all women and men to affordable and quality technical, vocational, and tertiary education, including university.                                                                                                   | <p><b>SDG Indicator</b> 4.3.1 Participation rate of youth and adults in formal and non-formal education and training in the previous 12 months, by gender.</p> <p><b>Monitoring Parameter:</b> Number of people receiving formal and/or non-formal education and training.</p>                                                                                 |
| 5 - Gender Equality             | <p><b>Social Sustainability:</b> The project ICS shall save time in cooking activities (by faster cooking) as well as save time in fuel procurement (reduced collection time or reduced fuel purchase trips to market) due to reduced fuel consumption.</p> <p>This shall save time spent by the women involved in these activities (unpaid work) to be utilized in the same in more productive and income generating tasks. The project will lead to helping beneficiary households in saving time in cooking activities (by faster cooking) as well as fuel procurement (reduced collection time or reduced fuel purchase trips to market).</p> | 5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate. | <p><b>SDG Indicator</b> 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location.</p> <p><b>Monitoring Parameter:</b> The average time saving associated with cooking and fuel collection time due to adoption of project technology / measures.</p>                                                                          |
| 7 - Affordable and Clean Energy | <p><b>Social, Economic Sustainability:</b> The project activity by distributing ICS will provide access to clean cooking with primary reliance on clean fuels and technology. The project ICS will be made available at a subsidized cost to beneficiary households. The project will lead to project households getting access to clean cookstoves (ICS) by the end of its lifetime.</p>                                                                                                                                                                                                                                                         | 7.1 By 2030, ensure universal access to affordable, reliable, and modern energy services.                                                                                                                                                               | <p><b>SDG Indicator</b> 7.1.2 Proportion of population with primary reliance on clean fuels and technology.</p> <p><b>Monitoring Parameter:</b></p> <ol style="list-style-type: none"> <li>1. Number of Beneficiaries under the project i.e., Number of ICS distributed under the project.</li> <li>2. % Users reporting operational project stove.</li> </ol> |
| 8 - Decent Work and             | <p><b>Economic Sustainability:</b> The project will create local employment opportunities in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.5 By 2030, achieve full and productive employment and decent work for all women                                                                                                                                                                       | <p><b>SDG Indicator</b> 8.5.1 Average hourly earnings of employees, by sex,</p>                                                                                                                                                                                                                                                                                |

|                     |                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |                                                                                                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Economic Growth     | operational and management roles, as well as in manufacturing, distribution and logistics.                                                                                                                                               | and men, including for young people and persons with disabilities, and equal pay for work of equal value.                                                                                                     | age, occupation and persons with disabilities<br><b>Monitoring Parameter:</b> Number of male / female employment created by project.                                              |
| 13 - Climate Action | <b>Environmental Sustainability:</b> The project by replacing the inefficient traditional stove with the energy-efficient ICS will reduce GHG emissions.                                                                                 | 13.2 Integrate climate change measures into national policies, strategies, and planning.                                                                                                                      | <b>SDG Indicator 13.2.2</b> Total greenhouse gas emissions per year<br><b>Monitoring Parameter:</b> Amount of CO <sub>2</sub> e emissions avoided or sequestered.                 |
| 15 - Life on Land   | <b>Environmental Sustainability:</b> The project will help reduce the use of non-renewable biomass from forests, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats. | 15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally. | <b>SDG Indicator 15.2.1</b> Progress towards sustainable forest management.<br><b>Monitoring Parameter:</b> Total non-renewable wood fuel saved after shifting to project device. |

## 1.18 Additional Information Relevant to the Project

### Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

### Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

### Further Information

Not Applicable.

## 2 SAFEGUARDS

### 2.1 No Net Harm

No potential negative socio-economic impacts are generated by the projects. The table below outlines various aspects of the project, to assess if any negative impacts exist and its mitigation measures, if required.



| Requirement Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Relevance (Yes/Potentially/No) | How the Project will Achieve Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mitigation Measures (if required)                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>1. Human Rights</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |
| 1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Yes                            | The Project activity and PP both respect human rights and are not complicit in violence or human rights abuses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | No negative impacts, hence, mitigation measures are not Applicable |
| 2. The Project shall not discriminate with regards to participation and inclusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Yes                            | The Project activity does not discriminate with regards to participation and inclusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No negative impacts, hence, mitigation measures are not Applicable |
| <b>2. Gender Equity</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |
| 1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women <ul style="list-style-type: none"> <li>a. Sexual harassment and/or any forms of violence against women – address the multiple risks of gender-based violence, including sexual exploitation or human trafficking.</li> <li>b. Slavery, imprisonment, physical and mental drudgery, punishment or coercion of women and girls.</li> <li>c. Restriction of women's rights or access to resources (natural or economic).</li> <li>d. Recognise women's ownership rights regardless of marital status – adopt project measures where possible to support to women's access to inherit and own land,</li> </ul> | Yes                            | <ul style="list-style-type: none"> <li>1. The primary beneficiaries of the Project activity, include domestic households, where women usually serve as primary cook. Thus, ICSs shall contribute towards Gender equality by freeing the women from the drudgery of fuel collection and cooking. The time saved in these activities would also help them to conduct other more productive /income generation activities. This would further contribute towards Gender Equality.</li> <li>2. PP does not discriminate on the basis on gender and promotes equal work for equal pay. PP further confirms that the average wages /</li> </ul> | No negative impacts, hence, mitigation measures are not Applicable |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>homes, and other assets or natural resources.</p> <p>2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work</p> <p>a. Where appropriate for the implementation of the project, paid, volunteer work or community contributions will be organised to provide the conditions for equitable participation of men and women in the identified tasks/activities.</p> <p>b. Introduce conditions that ensure the participation of women or men in Project activities and benefits based on pregnancy, maternity/paternity leave, or marital status.</p> <p>c. Ensure that these conditions do not limit the access of women or men, as the case may be, to Project activity participation and benefits.</p> <p>3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks</p> <p>4. (Where required) Summary of opinions and recommendations of an Expert Stakeholder(s)</p> |     | <p>remunerations paid to the employees working with PP is at / above par with the industry standards and in compliance with the minimum wage rate prescribed by the state / local government regulations.</p> <p>3. The project complies with National Women Development &amp; Gender Policy and does not limit women access to or control of resources, entitlements, and benefits. On the contrary, the PA ameliorates the living condition of women by freeing them from the burden of biomass fuel collection, decreasing smoke levels and incidence of respiratory illness relative to cooking on traditional biomass stoves using solid biomass fuel.</p> <p>4. Not Required</p> |  |
| <b>3. Community Health, Safety and Working Conditions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| 1. The Project shall avoid community exposure to increased health risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Yes | The Project activity reduces exposure to indoor air pollutants and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |

|                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| and shall not adversely affect the health of the workers and the community                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                     | smoke levels, further reducing incidence of respiratory illness compared to cooking on traditional biomass stoves using solid biomass fuel. |                                                                    |
| <b>4.1 Sites of Cultural and Historical Heritage</b>                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                    |
| 1. Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional, or religious values or intangible forms of culture?                                                                                               | No, the ICSs are installed in the kitchen of beneficiary households. The ICSs do not interfere with any sites, structures, or objects with historical, cultural, artistic, traditional, or religious values or intangible forms of culture                                                                                                                          | Requirements do not apply                                                                                                                   | No negative impacts, hence, mitigation measures are not Applicable |
| <b>4.2 Forced Eviction and Displacement</b>                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                    |
| 1. Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?                                                                                                                                     | No, the project ICS are installed in the kitchens of the household and hence any possibilities of forced eviction/physical relocation of people due to the project is not applicable. The PP collaborates with the local entrepreneurs for fabrication as well as installation of project ICS thereby empowering them economically rather than relocation / threat. | Requirements do not apply                                                                                                                   | No negative impacts, hence, mitigation measures are not Applicable |
| <b>4.3 Land Tenure and Other Rights</b>                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                    |
| 1. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?<br>2. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, | No, refer justification for 4.1,4.2 above.                                                                                                                                                                                                                                                                                                                          | Requirements do not apply                                                                                                                   | No negative impacts, hence, mitigation measures are not Applicable |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| access rights, usage rights or land ownership?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |
| <b>4.4 Native People</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |
| 1. Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | No, the project does not involve any land/territory claim. The ICS are installed in the project beneficiary households at their discretion. | Requirements do not apply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | No negative impacts, hence, mitigation measures are not Applicable |
| <b>5. Corruption</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |
| 1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Yes                                                                                                                                         | The PP does not promote / or is complicit in direct or indirect corruption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | No negative impacts, hence, mitigation measures are not Applicable |
| <b>6.1 Labour Rights</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |
| 1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions<br>2. Workers shall be able to establish and join labour organisations<br>3. Working agreements with all individual workers shall be documented and implemented and include: <ol style="list-style-type: none"> <li>Working hours (must not exceed 48 hours per week on a regular basis), AND</li> <li>Duties and tasks, AND</li> <li>Remuneration (must include provision for payment of overtime), AND</li> <li>Modalities on health insurance, AND</li> <li>Modalities on termination of the contract with provision for</li> </ol> | Yes                                                                                                                                         | 1. The Project activity does not involve any forced labour and the PP ensures that all employment is in compliance with local labour regulations and laws.<br>2. The PP puts no constraints / limitations on employees to form a union.<br>3. The CME/PI's policies (i.e. "Safeguarding Policy", "Inclusivity, Non-discrimination & Harassment Policy", "Child Labour Policy", "Code of Business Ethics Policy", "Anti-Corruption policy", etc. and employment contracts are compliant with the requirement<br>4. The PP does not promote / or is | No negative impacts, hence, mitigation measures are not Applicable |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <p>voluntary resignation by employee, AND</p> <p>f. Provision for annual leave of not less than 10 days per year, not including sick and casual leave.</p> <p>4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)</p> <p>5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures</p> |                                                                                                                                                                                                                                  | <p>complicit in child labour</p> <p>5. No, Major occupational hazards are envisaged during the operation of the project activity.</p> |                                                                           |
| <b>6.2 Negative Economic Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                           |
| <p>1. Does the project cause negative economic consequences during and after project implementation?</p>                                                                                                                                                                                                                                                                                                                                                                                                          | <p>No, the project will rather contribute to money and time saving because of the reduced fuel consumption. Therefore, the economic outcomes are expected to be positive.</p>                                                    | <p>Requirements do not apply</p>                                                                                                      | <p>No negative impacts, hence, mitigation measures are not Applicable</p> |
| <b>7.1 Emissions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                           |
| <p>1. Will the Project increase greenhouse gas emissions over the Baseline Scenario?</p>                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>No, the project shall reduce GHG emission by virtue of the increased efficiency of the ICS.</p>                                                                                                                               | <p>Requirements do not apply</p>                                                                                                      | <p>No negative impacts, hence, mitigation measures are not Applicable</p> |
| <b>7.2 Energy Resource</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                           |
| <p>1. Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?</p>                                                                                                                                                                                                                                                                                                        | <p>No, the project does not use or intervene with the fuel type used in baseline scenario. In fact, the project minimizes the consumption and thereby reduces the emission by virtue of the increased efficiency of the ICS.</p> | <p>Requirements do not apply</p>                                                                                                      | <p>No negative impacts, hence, mitigation measures are not Applicable</p> |
| <b>8.1 Impact on Natural Water Patterns/Flows</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                           |

|                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------|
| 1. Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?                           | No, the by-products (ash and smoke) of the project cookstoves operation are same as baseline scenario since there is no fuel change.<br><br>This rule out any additional potential impact on natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity? | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>8.2 Erosion and/or Water Body Instability</b>                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                                                    |
| 1. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?<br><br>2. Is the Project's area of influence susceptible to excessive erosion and/or water body instability? | No, The Project activity shall result in reduction in demand of biomass fuel in the region putting less pressure on forests for deforestation and will hence indirectly avoid erosion associated with tree cutting/felling.                                                                                                                                                                   | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.1 Landscape Modification and Soil</b>                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                                                    |
| 1. Does the Project involve the use of land and soil for production of crops or other products?                                                                                                                                                            | No, project ICSs shall be installed in the kitchen/cooking area of the beneficiary.                                                                                                                                                                                                                                                                                                           | Requirements do not apply | Not Applicable                                                     |
| <b>9.2 Vulnerability to Natural Disaster</b>                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                                                    |
| 1. Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought, or other extreme climatic conditions?                                                                   | No such consequences are envisaged.                                                                                                                                                                                                                                                                                                                                                           | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.3 Genetic Resources</b>                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                                                    |
| 1. Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g.,                                                                                                                                                    | No, the project intervention includes replacement of baseline stoves with improved project ICS.                                                                                                                                                                                                                                                                                               | Requirements do not apply | No negative impacts, hence, mitigation                             |

|                                                                                                                                                                 |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------|
| contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)? | It does not carry out any change in baseline fuel or that may add additional pollutant to in the air.                                                                                                                                                                            |                           | measures are not Applicable                                        |
| <b>9.4 Release of pollutants</b>                                                                                                                                |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
| 1. Could the Project potentially result in the release of pollutants to the environment?                                                                        | No, the project intervention includes replacement of baseline stoves with improved project ICS. It does not carry out any change in baseline fuel or that may add additional pollutant to in the air.                                                                            | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.5 Hazardous and Non-hazardous Waste</b>                                                                                                                    |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
| 1. Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?                             | No, the Project does not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?                                                                                                                                         | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.6 Pesticides &amp; Fertilisers</b>                                                                                                                         |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
| 1. Will the Project involve the application of pesticides and/or fertilisers?                                                                                   | No, the Project does not involve the application of pesticides and/or fertilisers                                                                                                                                                                                                | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.7 Harvesting of Forests</b>                                                                                                                                |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
| 1. Will the Project involve the harvesting of forests?                                                                                                          | No, The Project activity does not involve harvesting of forests. The Project shall result in reduction in demand of biomass fuel in the region putting less pressure on forests for deforestation and will hence indirectly avoid erosion associated with tree cutting/ felling. | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.8 Food</b>                                                                                                                                                 |                                                                                                                                                                                                                                                                                  |                           |                                                                    |
| 1. Does the Project modify the quantity or nutritional quality of food available such as through crop                                                           | No, the project does not intervene with the baseline food or other edible items.                                                                                                                                                                                                 | Requirements do not apply | No negative impacts, hence, mitigation                             |

|                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------|
| regime alteration or export or economic incentives?                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           | measures are not Applicable                                        |
| <b>9.9 Animal Raising</b>                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                                                    |
| 1. Will the Project involve animal husbandry?                                                                                                                                                                                                                                                              | No, the project will not involve animal husbandry                                                                                                                                                                                                                                                                                                                                                                                                               | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.10 High Conservation Value Areas and Critical Habitation</b>                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                                                    |
| 1. Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?                                                                                                                      | <p>No, the ICS are going to decrease the GHG emission levels solely due to improved performance and design of the project ICSSs. It does not involve change in baseline fuel or any other major interventions that may add additional pollutant to in the air.</p> <p>Thus, the project does not affect or alter the largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified.</p> | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |
| <b>9.11 Threatened Species</b>                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                                                    |
| <p>Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?</p> <p>and/or</p> <p>Does the Project potentially impact other areas where endangered species may be present through transboundary effects?</p> | <p>No, the ICS are going to decrease the GHG emission levels solely due to improved performance and design of the project ICSSs. It does not involve change in baseline fuel or does any other major interventions that may add additional pollutant to in the air.</p> <p>Therefore, the project implementation does not pose any threat to the endangered</p>                                                                                                 | Requirements do not apply | No negative impacts, hence, mitigation measures are not Applicable |



|  |                                 |  |  |
|--|---------------------------------|--|--|
|  | species prevalent in that area. |  |  |
|--|---------------------------------|--|--|

## 2.2 Local Stakeholder Consultation

The local stakeholder consultation (LSC) was conducted on 13 December 2022 in Kenya as follows:

|                   |                                                       |
|-------------------|-------------------------------------------------------|
| Date of LSC       | 13 December 2022                                      |
| Time (Local Time) | 10:00 am – 1:00 pm                                    |
| Venue             | The Heron Portico hotel, Jakaya Kikwete Road, Nairobi |
| Attendance        | 54 Physical, 23 Virtual                               |

The stakeholders were invited through various means such as email invitations, newspaper advertisement, invitation letters, phone calls, social media ads, posters advertisement at public places, etc. to collect their feedback on the project.

Email invitations were sent to stakeholders (Local Communities, Government officials, NGOs, local association leaders etc.) on 01/12/2022 - 02/12/2022 along with the non-technical summary of the project and stakeholder feedback forms. The Stakeholders were allowed to submit their feedback within 2 weeks from the date of receipt of the email via the feedback form provided.

The meeting had the presence of all categories of stakeholders targeted, with the consultation session being quite productive and engaging. Stakeholders present during meeting were provided with a non-technical summary of the project and a brief about the project and its components (including costs, risks, and benefits). The representatives of d.light and Climate Secure were also present to address any questions/queries from the participating stakeholders.

During the meeting, important aspects of the proposed project activity were discussed including technology/measure, baseline, grievance mechanism, SDG benefits, transfer of ownership of credits etc. Stakeholder feedbacks were collected via question-and-answer session during the consultation as well as a feedback form designed specifically to collect stakeholder opinion and rating on various aspects of the project.

As a part of on-going communication with local stakeholders, end users were informed about the grievance register which is maintained at the local office locations of the PP. Thus, it was explained during the meeting that end users/stakeholders are free to voice grievances regarding the project activity in the grievance register. Any relevant concern received during the operation of project activity, will be addressed according to it's merit.



Stakeholder Consultation meeting event (13 December 2022)



In general, the participating stakeholders were appreciative of the initiative taken by d.light and welcomed the project. Stakeholders acknowledged the features of project i.e., the reduction in Indoor air pollution / GHG and employment generation features of the project. No negative comments were raised during the Question session and feedback round.

Question/concerns raised during the physical stakeholder event by the participating stakeholders, and answers provided are given below for reference:

| Question/ Concern Raised                                                                                                                       | Response Provided                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How vulnerable households can acquire improved cookstoves and where are the cookstoves manufactured?                                           | The cookstoves are being manufactured locally or may be brought from outside neighbouring countries. The stove prices are subsidized to make it affordable to vulnerable households. It will be better to cover the logistics cost to acquire improved cookstoves, because if the consumer acquires the stoves by paying some money, he/she will definitely use it. |
| Who's the actual owner of the cookstoves and who owns the carbon credits?                                                                      | Customers are the sole owner of the cookstoves and d.light owns the carbon credits that are being generated by the usage of ICS. d.light will also be ensuring quality control, monitoring, training, guiding, and providing instructions to local sales and distribution partners. The improved cookstoves are provided by d.light at subsidized cost.             |
| How Improved Cookstoves are regionally distributed?                                                                                            | d.light has mapped and identified different regions which uses a lot of charcoal and firewood and know where the demand is. The plan is to distribute the cookstoves where the demand is through mapping and baseline survey.                                                                                                                                       |
| Are there any extensional services like training people on better ways using improved cookstoves in rural areas because it can be complicated. | There's basic Dos and Don'ts on how to use the cookstoves like by protect it from moisture, not washing the stoves immediately after use etc. There's no complexity in use of this improved cookstoves or problems in the use of these improved cookstoves.                                                                                                         |
| A member of NGO Gold Kenya, had a concern if the project is more focused on carbon credits or Biodiversity.                                    | The Improved Cookstoves have a more social impact and the distribution of the cookstoves is highly positive. Explanation was provided regarding how the improved cookstoves is acquired from d.light and the ownership of carbon credits as d.light doesn't generate revenue from the sales of ICS.                                                                 |
| Suggestion- Can the cook stoves use plastic as a source of energy since it's not biodegradable.                                                | d.light will not promote burning of plastics and these cookstoves are not designed in using plastics, since plastics are hazardous to health. The plastics is used if and only if there's lack of affordability of other traditional conventional fuels like charcoal and wood.                                                                                     |

Summary of feedback received via "stakeholder feedback form" during the physical meeting or virtually from invited stakeholders are given below:

### 1. Stakeholder Feedback - General

| S. No. | Question                                                                             | Summary of Feedback received                                                                                                  |
|--------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1      | What are your views on system of waiver of ownership of carbon credits by end users? | The stakeholders were positive regarding the system of waivership of ownership of carbon credits by end users to d.light. The |

|   |                                                                                                                                                              |                                                                                                                                                                                                                                              |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                                                              | stakeholders acknowledged that this will help future expansion of the project.                                                                                                                                                               |
| 2 | Kindly express your views on the Continuous grievance / feedback mechanism explained above?                                                                  | The stakeholders expressed that grievance mechanism will ensure efficiency in communication. The stakeholders also encouraged use of additional modes of communication like SMS, WhatsApp, social media etc.                                 |
| 3 | What recommendations do you have for monitoring procedures discussed above regarding the SDG indicators?                                                     | The stakeholders liked the monitoring parameters and their relevance to the project activity. The stakeholder supported determination of SDG impact via collection of feedback from customer using questionnaire survey in the project area. |
| 4 | What recommendations do you have for betterment of the project (improvement in implementation framework, interaction with other initiatives in region etc.)? | The stakeholders wished for the project to reach out all over Kenya (primarily rural areas) and provide education/training on usage and importance of clean cooking.                                                                         |
| 5 | Any other comment (technology, geographical spread/location etc.)                                                                                            | Stakeholders recommended the project to spread across rural areas.                                                                                                                                                                           |

## 2. Local Stakeholder Consultation – Overall Evaluation

|                                                                           |                                                                                                                                                                                                     |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question:</b> What is your impression of the Stakeholder Consultation? | The meeting was regarded as being very informative, progressive, interactive and productive.                                                                                                        |
| <b>Question:</b> What do you like about the project?                      | Project being Sustainable, generating employment opportunity and resulting in environment preservation were appreciated by stakeholders besides acknowledging the stoves to be affordable and nice. |
| <b>Question:</b> What do you not like about the project?                  | No significant concerns related to project design or technology were raised.                                                                                                                        |

### 2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

### 2.4 Public Comments

The project was opened for public consultation from 04/04/2023-04/05/2023. No feedback was received during the public comment period.

### 2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

## 3 APPLICATION OF METHODOLOGY

### 3.1 Title and Reference of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 dated 22 July 2021

<https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

For calculation of fraction of non-renewable biomass, the below tool is used “TOOL30: Calculation of the fraction of non-renewable biomass”,

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

For demonstration of Additionality- “Tool 1: Tool for demonstration and assessment of additionality”, Version-7.0.0”:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

Tool 27: Investment analysis, Version- 11.0.

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf>

### 3.2 Applicability of Methodology

The applicability of VMR0006 methodology is justified as below:

The applicability of Vintec's methodology is justified as below:

| Applicability criterion                                                                                                                                                                                | Project Activity Applicability                                                                                                                                                                                                                                                                                                                                                                    |                                  |                      |         |  |  |      |      |      |      |        |         |         |         |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------|---------|--|--|------|------|------|------|--------|---------|---------|---------|---------|
| Project activities shall be implemented in domestic premises, or in community-based kitchens                                                                                                           | The proposed project involves deployment of ICS in domestic households and communities in Kenya.                                                                                                                                                                                                                                                                                                  |                                  |                      |         |  |  |      |      |      |      |        |         |         |         |         |
| The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; | ICS planned to be distributed under the project are energy efficiency cookstove using woody biomass as fuel and have an efficiency more than 25%.                                                                                                                                                                                                                                                 |                                  |                      |         |  |  |      |      |      |      |        |         |         |         |         |
| Both 'Projects' and 'Large Projects' can use this methodology                                                                                                                                          | The proposed project is a large-scale project as the annual emission reduction is more than 300,000 tonnes of CO <sub>2</sub> e per year which is in line with the section 3.10.1 of VCS standard version 4.4.                                                                                                                                                                                    |                                  |                      |         |  |  |      |      |      |      |        |         |         |         |         |
| Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports, or statistics                           | Non-renewable biomass has been in use since 31 December 1989 as evidenced by various FAO statistical data given below. <table><tr><th rowspan="2">FRA 2020 categories<sup>7</sup></th><th colspan="4">Area (1000 hectares)</th></tr><tr><th>1990</th><th>2000</th><th>2010</th><th>2020</th></tr><tr><td>Forest</td><td>3858.54</td><td>3961.19</td><td>3616.34</td><td>3611.09</td></tr></table> | FRA 2020 categories <sup>7</sup> | Area (1000 hectares) |         |  |  | 1990 | 2000 | 2010 | 2020 | Forest | 3858.54 | 3961.19 | 3616.34 | 3611.09 |
| FRA 2020 categories <sup>7</sup>                                                                                                                                                                       | Area (1000 hectares)                                                                                                                                                                                                                                                                                                                                                                              |                                  |                      |         |  |  |      |      |      |      |        |         |         |         |         |
|                                                                                                                                                                                                        | 1990                                                                                                                                                                                                                                                                                                                                                                                              | 2000                             | 2010                 | 2020    |  |  |      |      |      |      |        |         |         |         |         |
| Forest                                                                                                                                                                                                 | 3858.54                                                                                                                                                                                                                                                                                                                                                                                           | 3961.19                          | 3616.34              | 3611.09 |  |  |      |      |      |      |        |         |         |         |         |

<sup>7</sup> <https://www.fao.org/3/cb0019en/cb0019en.pdf>

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                 |                 |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Other wooded land                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 34850.25        | 33240.61        | 33099.15        | 32271.45        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Other land                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 18205.21        | 19712.20        | 20198.51        | 21031.46        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Total Land</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>56914.00</b> | <b>56914.00</b> | <b>56914.00</b> | <b>56914.00</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Kenya lost an average of 8,248 hectares of forest per year with average annual deforestation rate of 0.21% between 1990 and 2010. In total, Kenya lost 6.4% of its forest or 247,450 hectares.                                                                                                                                                                                                                                                                                                                                                                                      |                 |                 |                 |                 |
| For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips), it shall be demonstrated that:                                                                                                                                                                                                                                                                                                                                                                                     | <p>The project does not envisage use of processed biomass. In the case where processed biomass residues are used in the project, the following will be ensured:</p> <ol style="list-style-type: none"> <li>1. It is renewable biomass.</li> <li>2. Its consumption will be monitored.</li> <li>3. If renewable biomass processing (e.g., shredding and compacting in the case of briquetting) occurs in project case, the energy consumption for manufacturing and transportation of renewal biomass fuel (briquettes) will be monitored to calculate project emissions.</li> </ol> |                 |                 |                 |                 |
| <ol style="list-style-type: none"> <li>1. It is produced using exclusively renewable biomass (more than one type of biomass may be used).</li> <li>2. The consumption of the fuel should be monitored during the crediting period and</li> <li>3. Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded</li> </ol> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                 |                 |                 |
| The VCS PD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).                                                                                                                                                                                                                                                                   | <p>The ICS installation database will have the following provisions to ensure complete avoidance of double counting:</p> <ol style="list-style-type: none"> <li>1. Unique serial number of ICS distributed.</li> <li>2. End users' details (End username, address, and/or telephone number (if available))</li> </ol>                                                                                                                                                                                                                                                               |                 |                 |                 |                 |
| The VCS PD shall also explain how the proposed procedures prevent double counting of                                                                                                                                                                                                                                                                                                                                                                                                                             | At the time of ICS distribution, each ICS beneficiary transfers the rights of credit ownership to PP via an agreement or disclaimed via clear statement on the product packing or                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                 |                 |                 |

|                                                                                                                                                                          |                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices. | warranty card thereby avoiding any subsequent double claim on ownership of credits by other entities. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|

The applicability of tool 01 is justified as below:

| Applicability criterion                                                                                                                                                                                                                                                                                                                                                                                | Project Activity Applicability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Condition para 9:</b><br>The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool. | Not applicable, an approved methodology is being applied in the project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Condition para 10:</b><br>Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.                                                                                                                                                                                                                           | Step-3 in Section 7 of the applied methodology VMR 0006, version 1.1 states the following:<br><i>“For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible.”</i><br>Since the proposed project does not distribute ICS at zero cost, hence the Tool 1 is applicable to the project. |

The applicability of tool 27 is justified as below:

| Applicability criterion                                                                                                                                                                                                                                                                                                                                                                                                                                               | Project Activity Applicability                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Condition para 2:</b><br>This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the | As project applies “Tool for the demonstration and assessment of additionality”. Thus, TOOL 27 can be applied to the project. Please refer to section B.5 of PD for details. |

|                                                                                                                                                                                                                                                                                      |                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.                                                                                                                                                                       |                                                                               |
| <b>Condition para 3:</b><br><br>In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. | The applied methodology does not define any investment analysis requirements. |

The applicability of tool 30 is justified as below:

| Applicability criterion                                                                                                                                                                                                                                                                                                                                   | Project Activity Applicability                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| This tool may be used by: <ul style="list-style-type: none"> <li>a) DNAs to submit region/country-specific default fNRB values, following the procedures for development, revision, clarification, and update of standardized baselines (SB procedures); or</li> <li>b) project participants to calculate project or PoA-specific fNRB values.</li> </ul> | This tool is used by the project participants to calculate a project specific value for fNRB under option (b) of applicability condition 1. |
| For project or PoA specific fNRB values, project participants shall assess the area where biomass is sourced for end-users included in the project activity and justify the selection of the area in project design documents.                                                                                                                            | The area for the calculated value is selected within the geographical boundary of Project.                                                  |

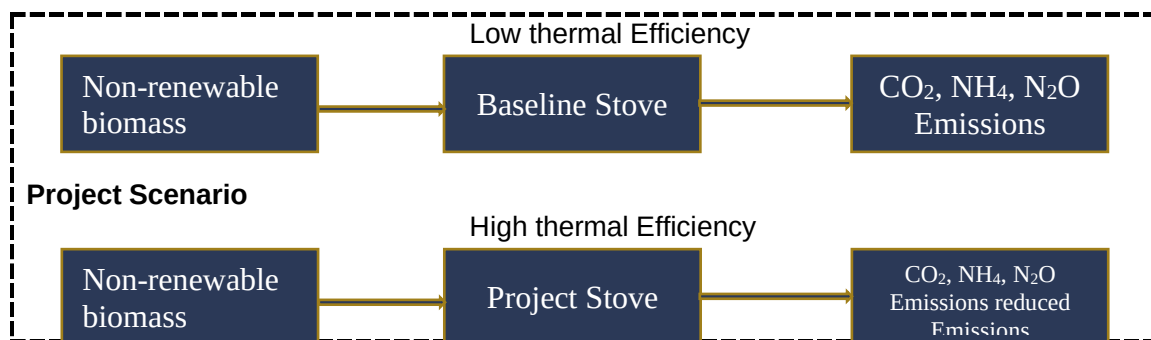
### 3.3 Project Boundary

As per the applied methodology, the project boundary is the physical and geographical location of all equipment and systems affected by the project activity. As the activity involves distribution of ICS (Improved Cookstove), the project boundary is the physical, geographical location of each household where the ICS (Improved Cookstove) is distributed to replace the traditional stove.

| Source   |                                                        | Gas              | Included? | Justification/Explanation  |
|----------|--------------------------------------------------------|------------------|-----------|----------------------------|
| Baseline | Emission from use of non-renewable biomass/Fossil fuel | CO <sub>2</sub>  | Yes       | Major Source               |
|          |                                                        | CH <sub>4</sub>  | Yes       | Major Source               |
|          |                                                        | N <sub>2</sub> O | Yes       | Major Source               |
|          |                                                        | Other            | No        | No other source identified |
| Project  | Emission from use of non-renewable biomass/Fossil fuel | CO <sub>2</sub>  | Yes       | Major Source               |
|          |                                                        | CH <sub>4</sub>  | Yes       | Major Source               |
|          |                                                        | N <sub>2</sub> O | Yes       | Major Source               |
|          |                                                        | Other            | No        | No other source identified |

A representation of baseline and project boundary are given as below:

|                                                          |
|----------------------------------------------------------|
| <b>Republic of Kenya</b><br><br><b>Baseline Scenario</b> |
|----------------------------------------------------------|



### 3.4 Baseline Scenario

As per section 6 of the methodology VMR0006 Ver 1.1, "The baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cook stoves in absence of project activity."

Refer section 1.11 above, wherein the prevailing practice in Kenya has been substantiated as use of non-renewable wood fuel (such as firewood or charcoal) for traditional cooking methods using national government literature / publications.

### 3.5 Additionality

The additionality has been assessed in line with section 7.0 of the applied methodology.

#### Step 1: Regulatory Surplus

The proposed project is a voluntary coordinated action by d.light, and there is no mandated government programme or policy in the host country of this project ensuring the distribution of fuel-efficient cookstoves. The project is not mandated by any law, statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework (refer section 1.14 above).

#### Step 2: Positive List

Under the proposed project, project ICS are not being distributed at zero costs but shall be distributed at a subsidized price to end users. Hence step 2 is not applicable for demonstration of project's additionality.

#### Step 3: Project Method

Investment analysis method as set out in the "Tool 1: Tool for demonstration and assessment of additionality", Version-7.0.0 and "Tool 27: Investment Analysis, version 11.0, has been applied to demonstrate the additionality as follows:

#### **Step 2.** Investment analysis of the "Tool for demonstration and assessment of additionality" Version-7.0.0"

The MOU was signed between d.light (PP) and the ICS supplier on June 14, 2022. This specific date has been considered as the official investment decision date as this marks first point of no return from the project and PP's commitment to investment in the project.

#### **Sub-step 2a:** Determine appropriate analysis method.

Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis (Sub-step 2b). If the CDM project activity and the alternatives identified in Step 1 generate no



financial or economic benefits other than CDM related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

The project will distribute ICS at a subsidized price (i.e., at a price lower than the purchase cost of ICS). There is no financial or economic benefit other than VER income, given all sale revenues are less than the cost to PP per unit of ICS<sup>8</sup>. Although simple cost analysis can be applied, a better presentation might be achieved via Investment Comparison Analysis.

#### Sub-step 2b: Option II. Apply Investment Comparison Analysis

Identify the financial indicator, such as IRR, NPV, cost benefit ratio, or unit cost of service (e.g., levelized cost of electricity production in \$/kWh or levelized cost of delivered heat in \$/GJ) most suitable for the project type and decision-making context.

The project applies NPV analysis. The project's NPV is negative without VER revenue and hence substantiates that the project activity is not financially viable / economically attractive without the sale of VER revenues.

The following table presents the costs associated with project stove distribution.

| Category | Input Values                                        | Value (KSH)              | Comment                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COSTS    | <b>ICS Purchase Cost from Manufacturer per unit</b> | <b>2,460.52</b>          | Agreement with manufacturer, page 5, Appendix 1                                                                                                                                                                                                                                                                               |
|          | <b>Transport &amp; Storage Cost per unit</b>        | <b>297.91</b>            | Stocking commissions evidence paid out per unit and Dispatch agreement                                                                                                                                                                                                                                                        |
|          | Stocking Commission per unit                        | 150.00                   |                                                                                                                                                                                                                                                                                                                               |
|          | Shipping Cost per unit                              | 17.91                    |                                                                                                                                                                                                                                                                                                                               |
|          | Dispatch Cost per unit                              | 130.00                   |                                                                                                                                                                                                                                                                                                                               |
|          | <b>Distribution Commission per unit</b>             | <b>725.00</b>            | Distribution commissions evidence paid out per unit                                                                                                                                                                                                                                                                           |
|          | Selling Commission to Sales agent per unit          | 600.00                   |                                                                                                                                                                                                                                                                                                                               |
|          | Fuel incentive to Sales agent per unit              | 125.00                   |                                                                                                                                                                                                                                                                                                                               |
|          | <b>Inflation rate</b>                               | <b>5.83%<sup>9</sup></b> | As per para 16 of Tool 27: Investment Analysis, v11.0. The value of inflation rate is derived from the average forecasted inflation rate for the host country (Kenya) over a 7-year period (from 2022 – 2028, i.e. duration of the crediting period), as provided by the International Monetary Fund's World Economic Outlook |
|          | <b>Tax (VAT)</b>                                    | <b>16.0%</b>             | <a href="https://www.kra.go.ke/individual/filing-paying/types-of-taxes/value-added-tax">https://www.kra.go.ke/individual/filing-paying/types-of-taxes/value-added-tax</a>                                                                                                                                                     |
|          | <b>Other Costs per unit</b>                         | <b>0.00</b>              |                                                                                                                                                                                                                                                                                                                               |

<sup>8</sup> The revenue from distribution of sale is not deemed to generate any economic or financial benefit, as the price at which stove is being provided to the beneficiary is less than the cost of stove procurement and distribution by PP.

<sup>9</sup> <https://www.imf.org/external/datamapper/PCPIPCH@WEO/OEMDC/ADVEC/WEOWORLD/>

|                      |                                                 |                          |                                                                                                                                                                                                                                      |
|----------------------|-------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | Stove Replacements costs due to warranty claims | 0.00                     | While these costs are significant (in the range of 2500-3000 (KSH) per ICS), these have been conservatively considered as 0, given these are internal costs and cannot be objectively justified via external / third party evidence. |
|                      | Operation and Maintenance costs                 | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Project Management costs                        | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Software development cost                       | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Call Centre costs                               | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Warehousing costs                               | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Marketing overheads                             | 0.00                     |                                                                                                                                                                                                                                      |
|                      | General business expenses                       | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Office related expenses                         | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Regional expenses                               | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Manpower costs                                  | 0.00                     |                                                                                                                                                                                                                                      |
|                      | Corporate overheads                             | 0.00                     |                                                                                                                                                                                                                                      |
|                      | <b>Total ICS Cost per unit</b>                  | <b>3,877.11</b>          |                                                                                                                                                                                                                                      |
| REVENUES             | <b>ICS sale price to beneficiary per unit</b>   | <b>1,990.00</b>          | Supply Agreement between d.light and Burn Manufacturing d.light and Burn contract                                                                                                                                                    |
|                      | <b>RBF revenue<sup>10</sup> per unit</b>        | <b>351.50</b>            |                                                                                                                                                                                                                                      |
| <b>Discount rate</b> |                                                 | <b>7.5%<sup>11</sup></b> | Consideration of central bank rate (Debt) is deemed most conservative as the discount rate.                                                                                                                                          |
| <b>NPV</b>           |                                                 | <b>Negative</b>          |                                                                                                                                                                                                                                      |

Despite considering a very conservative ICS cost above (with several costs considered as zero), the calculated value of the NPV is negative without VCUs revenue. This confirms that the project activity is not financially viable without the support of carbon revenues and is fully additional.

The alternative to the project activity is deemed as continuation of current practice by the potential beneficiaries i.e., use of traditional / inefficient means of cooking. For this alternative, the PP does not need to make any investment neither shall earn any revenue associated with the continuation of current practice. Thus, the NPV for the alternative is deemed as 0.

### Sensitivity Analysis:

The project activity's financial performance is dependent on several crucial parameters. This section does a sensitivity analysis to ensure that the financial performance is resilient to moderately favourable fluctuations in the essential assumptions. As per para 27 of Tool 27 Ver. 11 "Variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation".

The following set of conditions are expected to materially affect the NPV of the proposed project -

1. Increase in project revenue / increase in sale price of ICS

<sup>10</sup> Although the project has not availed any RBF benefit as of yet, this has been provisioned in the revenues as a conservative measure.

<sup>11</sup> <https://www.centralbank.go.ke/rates/central-bank-rate/>

2. Decrease in purchase cost of ICS
3. Decrease in other costs

The results of the sensitivity analysis are summarized below:

| Sensitivity Analysis        | NPV Achieved  | Variation needed to achieve '0' NPV |
|-----------------------------|---------------|-------------------------------------|
| revenue / sale price of ICS | - 297,201,689 | 71.20% increase                     |
| Purchase cost               |               | 56.50% decrease                     |
| Other costs                 |               | 157.64% decrease                    |

A ~71% increase in sale price of ICS is not possible. The target beneficiaries cannot afford to buy the ICS at such increased prices.

The purchase cost is binding as per the supply agreement with the ICS supplier. The purchase cost is likely to increase over time rather to decrease. Thus, it is not feasible to achieve a ~57% reduction in purchase cost of ICS.

Similarly, other costs have to reduce by more than 100% to make the NPV positive. Again, this is not feasible as for the same other costs have to become -ve which is not possible.

Thus, the project is not sensitive to significant variations to the input parameters and scenarios in which the project activity might become financially more favourable than the alternative does not exist. As per para 42a) of Tool1, v 7.0.0, since the NPV of alternative (continuation of current practice) is higher than the NPV of the proposed project without considering carbon revenues. hence, the project is substantiated as fully additional.

### 3.6 Methodology Deviations

The project does not involve any methodological deviations.

## 4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

### 4.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

### 4.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

### 4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

#### 4.4 Net GHG Emission Reductions and Removals

As per applicable methodology, formula to calculate emission reduction (ERs) is:

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad (1)$$

Where:

$i$  Indices for the situation where more than one type/model of improved cook stove is introduced to replace the baseline stove

$j$  Indices for the situation where there is more than one batch of improved cook stove of type  $i$

$ER_y$  Emission reductions during year  $y$  in t CO<sub>2</sub>e

$ER_{y,i,j}$  = Emission reductions by improved cook stove of type  $i$  and batch  $j$  during year  $y$  in t CO<sub>2</sub>e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad (2)$$

Where:

$B_{y,savings,i,j}$  Quantity of woody biomass that is saved in tonnes per improved cook stove of type  $i$  and batch  $j$  during year  $y$

$f_{NRB,y}$  Fraction of woody biomass that can be established as non-renewable biomass ( $f_{NRB}$ )

$NCV_{wood\ fuel}$  Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

$EF_{wf,CO_2}$  CO<sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO<sub>2</sub>/TJ)

$EF_{wf,non\ CO_2}$  Non-CO<sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO<sub>2</sub>/TJ)

$N_{y,i,j}$  Number of improved cook stoves of type  $i$  and batch  $j$  operating during year  $y$

0.95 Discount factor to account for leakage

The quantity of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left( \frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad (4)$$

Where:

$B_{y=1,new,i,survey}$  Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type  $i$  and batch  $j$ , determined in the first year of the implementation of the project through a sample survey.

$\eta_{old}$  Efficiency of baseline cookstove

$\eta_{new,i,j}$  Efficiency of the improved cook stove type  $i$  and batch  $j$  determined through water boiling test (WBT). efficiency may be determined using Equation 5.

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad (5)$$

Where:

$\eta_p$  Efficiency of project stove (fraction) at the start of project activity

$(DF_n)^{y-1}$  Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively, a default value of 0.99 efficiency loss per year can be considered.

0.94 Adjustment factor to account for uncertainty related to project cookstove efficiency test

#### fNRB Calculation

fNRB is calculated as per "TOOL30: Calculation of the fraction of non-renewable biomass", v 4.0.

For detailed calculation, refer fNRB calculator.

#### Ex-ante calculation of emission reduction

| Year         | Estimated baseline emissions or removals (tCO <sub>2</sub> e) | Estimated project emissions or removals (tCO <sub>2</sub> e) | Estimated leakage emissions (tCO <sub>2</sub> e) | Estimated net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|--------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|
| Year 1       | 365,221                                                       | 0                                                            | 0                                                | 365,221                                                                |
| Year 2       | 898,282                                                       | 0                                                            | 0                                                | 898,282                                                                |
| Year 3       | 883,659                                                       | 0                                                            | 0                                                | 883,659                                                                |
| Year 4       | 869,183                                                       | 0                                                            | 0                                                | 869,183                                                                |
| Year 5       | 854,851                                                       | 0                                                            | 0                                                | 854,851                                                                |
| Year 6       | 840,662                                                       | 0                                                            | 0                                                | 840,662                                                                |
| Year 7       | 826,615                                                       | 0                                                            | 0                                                | 826,615                                                                |
| <b>Total</b> | <b>5,538,473</b>                                              | <b>0</b>                                                     | <b>0</b>                                         | <b>5,538,473</b>                                                       |

## 5 MONITORING

### 5.1 Data and Parameters Available at Validation

The following parameters are fixed for the entire crediting period using default values as per the methodology:

$\eta_{new,i,j}$  Fixed ex-ante using default discount factor as per methodology in equation (5)

$\eta_{old}$  Fixed ex-ante using default values as per the methodology

$NCV_{biomass}$  Fixed ex-ante using default values as per the methodology

| Data / Parameter | $\eta_p$                                                     |
|------------------|--------------------------------------------------------------|
| Data unit        | Fraction                                                     |
| Description      | Efficiency of project stove at the start of project activity |

|                                                                                              |                                                                                                                                                    |                         |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Source of data                                                                               | Manufacturer's specification                                                                                                                       |                         |
| Value applied                                                                                | ICS Model                                                                                                                                          | Manufacturer Efficiency |
|                                                                                              | Kunikoa with pot-skirt                                                                                                                             | 51.3%                   |
|                                                                                              | Kunikoa without pot-skirt                                                                                                                          | 43.4%                   |
|                                                                                              | Ecoa Wood with pot-skirt                                                                                                                           | 51.3%                   |
|                                                                                              | Ecoa Wood without pot-skirt                                                                                                                        | 43%                     |
|                                                                                              | Jikokoa                                                                                                                                            | 53.7%                   |
|                                                                                              | Ecoa Char                                                                                                                                          | 53.7%                   |
| Justification of choice of data or description of measurement methods and procedures applied | As per the specification sheet provided by the manufacturer                                                                                        |                         |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                                                  |                         |
| Comments                                                                                     | In case of inclusion of other/new models, during project lifetime, corresponding thermal efficiency values shall be considered for ER calculations |                         |

|                                                                                              |                                                                                           |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $f_{NRB}$                                                                                 |
| Data unit                                                                                    | Fraction                                                                                  |
| Description                                                                                  | Fraction of woody biomass that can be established as non-renewable biomass                |
| Source of data                                                                               | Calculated                                                                                |
| Value applied                                                                                | 0.7875                                                                                    |
| Justification of choice of data or description of measurement methods and procedures applied | Calculated as per the "TOOL30: Calculation of the fraction of non-renewable biomass",v4.0 |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                         |
| Comments                                                                                     | Refer Appendix 2 for detail                                                               |

|                                                                                              |                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $NCV_{woodfuel}$                                                                                                                                                          |
| Data unit                                                                                    | TJ/tonne                                                                                                                                                                  |
| Description                                                                                  | Net calorific value of the non-renewable woody biomass                                                                                                                    |
| Source of data                                                                               | Default value (VMR0006 ver. 1.1, page 10)                                                                                                                                 |
| Value applied                                                                                | 0.0156                                                                                                                                                                    |
| Justification of choice of data or description of measurement methods and procedures applied | IPCC default for wood fuel, based on the gross weight of the wood that is 'air-dried', may be used if fuel used in the project device is woody biomass/renewable biomass. |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                                                                         |
| Comments                                                                                     | --                                                                                                                                                                        |

|                                                   |                                                                                                                  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                  | $EF_{wf,CO_2}$                                                                                                   |
| Data unit                                         | $tCO_2/TJ$                                                                                                       |
| Description                                       | $CO_2$ Emission factor for the fossil fuels projected to be used for substitution of non-renewable woody biomass |
| Source of data                                    | Default value (VMR0006 ver. 1.1, page 10)                                                                        |
| Value applied                                     | 112                                                                                                              |
| Justification of choice of data or description of | IPCC default value for wood fuel                                                                                 |

|                                            |                                   |
|--------------------------------------------|-----------------------------------|
| measurement methods and procedures applied |                                   |
| Purpose of Data                            | Calculation of baseline emissions |
| Comments                                   | --                                |

|                                                                                              |                                                                                                                           |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $EF_{wf,non\ CO_2}$                                                                                                       |
| Data unit                                                                                    | tCO <sub>2</sub> /TJ                                                                                                      |
| Description                                                                                  | CO <sub>2</sub> Emission factor for the fossil fuels projected to be used for substitution of non-renewable woody biomass |
| Source of data                                                                               | Default value (VMR0006 ver. 1.1, page 10)                                                                                 |
| Value applied                                                                                | 26.23                                                                                                                     |
| Justification of choice of data or description of measurement methods and procedures applied | IPCC default value for wood fuel                                                                                          |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                         |
| Comments                                                                                     | --                                                                                                                        |

|                                                                                              |                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\eta_{old}$                                                                                                                                                                                                                                                                                                             |
| Data unit                                                                                    | Fraction                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                  | Thermal efficiency of baseline cookstove                                                                                                                                                                                                                                                                                 |
| Source of data                                                                               | Default value (VMR0006 ver. 1.1)                                                                                                                                                                                                                                                                                         |
| Value applied                                                                                | 0.10 for Firewood<br>0.20 for Charcoal                                                                                                                                                                                                                                                                                   |
| Justification of choice of data or description of measurement methods and procedures applied | A default value of 0.1 shall be used if the baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is, without a grate or a chimney.<br><br>A default value of 0.2 shall be used for other types of devices. |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                                                                                                                                                                                                                        |
| Comments                                                                                     | As per the methodology, this value has been fixed for each household that receives project ICS, during the project crediting period. To be updated at renewal of crediting period                                                                                                                                        |

|                                                                                              |                                                                                                       |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $DF_n$                                                                                                |
| Data unit                                                                                    | Fraction                                                                                              |
| Description                                                                                  | Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). |
| Source of data                                                                               | Default value                                                                                         |
| Value applied                                                                                | 0.99                                                                                                  |
| Justification of choice of data or description of measurement methods and procedures applied | As per the applied methodology                                                                        |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                     |
| Comments                                                                                     | --                                                                                                    |

|                  |                  |
|------------------|------------------|
| Data / Parameter | $\eta_{new,i,j}$ |
|------------------|------------------|



| Data unit                                                                                    | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|-------|-------|-------|-------|---|---|--------------------------|-------|-------|-------|-------|-------|-------|-------|--------------------------|-------|-------|-------|-------|-------|-------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|----------------------|-------|-------|-------|-------|-------|-------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|
| Description                                                                                  | Efficiency of the device of each type i and batch j implemented as part of the project activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| Source of data                                                                               | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| Value applied                                                                                | As per equation (5) above the applicable efficiency of the example ICS models <sup>12</sup> to be used in Ex-post ER calculations is as follows:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
|                                                                                              | <table><tr><th><math>\eta_{new,i,j}</math></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th></tr><tr><td><math>\eta_{new,kuniokoa,ps}</math></td><td>48.22</td><td>47.74</td><td>47.26</td><td>46.79</td><td>46.32</td><td>45.86</td><td>45.40</td></tr><tr><td><math>\eta_{new,ecoawood,ps}</math></td><td>48.22</td><td>47.74</td><td>47.26</td><td>46.79</td><td>46.32</td><td>45.86</td><td>45.40</td></tr><tr><td><math>\eta_{new,kuniokoa}</math></td><td>40.80</td><td>40.39</td><td>39.98</td><td>39.58</td><td>39.19</td><td>38.80</td><td>38.41</td></tr><tr><td><math>\eta_{new,ecoawood}</math></td><td>40.42</td><td>40.02</td><td>39.62</td><td>39.22</td><td>38.83</td><td>38.44</td><td>38.05</td></tr><tr><td><math>\eta_{new,jikokoa}</math></td><td>50.48</td><td>49.97</td><td>49.47</td><td>48.98</td><td>48.49</td><td>48.00</td><td>47.52</td></tr><tr><td><math>\eta_{new,ecoachar}</math></td><td>50.48</td><td>49.97</td><td>49.47</td><td>48.98</td><td>48.49</td><td>48.00</td><td>47.52</td></tr></table> | $\eta_{new,i,j}$ | 1     | 2     | 3     | 4     | 5     | 6 | 7 | $\eta_{new,kuniokoa,ps}$ | 48.22 | 47.74 | 47.26 | 46.79 | 46.32 | 45.86 | 45.40 | $\eta_{new,ecoawood,ps}$ | 48.22 | 47.74 | 47.26 | 46.79 | 46.32 | 45.86 | 45.40 | $\eta_{new,kuniokoa}$ | 40.80 | 40.39 | 39.98 | 39.58 | 39.19 | 38.80 | 38.41 | $\eta_{new,ecoawood}$ | 40.42 | 40.02 | 39.62 | 39.22 | 38.83 | 38.44 | 38.05 | $\eta_{new,jikokoa}$ | 50.48 | 49.97 | 49.47 | 48.98 | 48.49 | 48.00 | 47.52 | $\eta_{new,ecoachar}$ | 50.48 | 49.97 | 49.47 | 48.98 | 48.49 | 48.00 | 47.52 |
| $\eta_{new,i,j}$                                                                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                | 3     | 4     | 5     | 6     | 7     |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,kuniokoa,ps}$                                                                     | 48.22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 47.74            | 47.26 | 46.79 | 46.32 | 45.86 | 45.40 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,ecoawood,ps}$                                                                     | 48.22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 47.74            | 47.26 | 46.79 | 46.32 | 45.86 | 45.40 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,kuniokoa}$                                                                        | 40.80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 40.39            | 39.98 | 39.58 | 39.19 | 38.80 | 38.41 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,ecoawood}$                                                                        | 40.42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 40.02            | 39.62 | 39.22 | 38.83 | 38.44 | 38.05 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,jikokoa}$                                                                         | 50.48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 49.97            | 49.47 | 48.98 | 48.49 | 48.00 | 47.52 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| $\eta_{new,ecoachar}$                                                                        | 50.48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 49.97            | 49.47 | 48.98 | 48.49 | 48.00 | 47.52 |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| Justification of choice of data or description of measurement methods and procedures applied | To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |
| Comments                                                                                     | In case of inclusion of other/new models, during the project lifetime, corresponding thermal efficiency values shall be used for Ex-post ER calculations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |       |       |       |       |       |   |   |                          |       |       |       |       |       |       |       |                          |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |                      |       |       |       |       |       |       |       |                       |       |       |       |       |       |       |       |

## 5.2 Data and Parameters Monitored

The following parameters as per section 9.2 of the methodology are deemed not applicable:

|                   |                                                                                |
|-------------------|--------------------------------------------------------------------------------|
| $\eta_{new,i,j}$  | Fixed ex-ante using default discount factor as per methodology in equation (5) |
| $\eta_{old}$      | Fixed ex-ante using default values as per the methodology                      |
| $B_{renewable,y}$ | Applicable only in case of use of equation (7)                                 |
| $\eta_{PJ,BL}$    | Applicable only in case of use of equation (7)                                 |
| $NCV_{biomass}$   | Fixed ex-ante using default values as per the methodology                      |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | $N_{y,i,j}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Data unit                                                       | Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Description                                                     | Number of project devices of type i and batch j operating during year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Source of data                                                  | Installation database and monitoring survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description of measurement methods and procedures to be applied | <p>CME shall maintain a database of all ICS distributed. The usage rate of installed ICS shall be measured based on a representative sample. CDM Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of CDM Standard for sampling and surveys for project activities and programme of activities.</p> <p>Alternatively, the simplified approach proposed in option (b), page 12 under Section 8.4 of the methodology may be used for determining the minimum sample size in which case compliance with 90/10 confidence precision is not obligatory.</p> |

<sup>12</sup>In case the project involves ICS models other than the one listed above; its corresponding thermal efficiency value calculated in accordance with equation (5) shall be used.

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequency of monitoring/recording | At least once every two years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Value applied                     | 200,000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Monitoring equipment              | Sampling based Monitoring survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| QA/QC procedures to be applied    | In the case the desired precision is not met, lower bound values shall be used against repeating the survey to determine the operational fraction of ICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Purpose of data                   | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Calculation method                | <p>The results from sample-based monitoring will be used to calculate <math>N_{y,i,j}</math> as follows</p> $N_{y,i,j} = ( n_{i,j,operational} / n_{i,j,total} ) * N_{y,i,j,distributed}$ <p>Where:<br/> N = number of stoves<br/> n = number of samples</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Comments                          | <p>During the stove distribution, the presence of pre-existing project ICS, if any, shall be monitored and in case a pre-existing project ICS is found in the same household, the subsequent (second) ICS will not be included in the ER calculations.</p> <p>Alternatively, the presence of multiple project ICS in a household may be determined ex-post during sampling surveys and the total ICS population shall be discounted by the fraction of sampled households found using more than one project ICS.</p> <p>Further, ex-post monitoring shall assess (on sampling basis) the continuity of usage of pot skirt by monitoring the usage of pot skirts. In case the project ICS that were distributed with pot skirts are found to be not using the same, the ERs shall be adjusted accordingly discounting for any drop off of pot skirts.</p> |

|                                                                 |                                                                                                                                                                                                        |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | Life Span                                                                                                                                                                                              |
| Data unit                                                       | Number of years                                                                                                                                                                                        |
| Description                                                     | Project promoters to state the operating lifetime of the project device for projects opting Equation 5 for determining project stove efficiency                                                        |
| Source of data                                                  | Manufacturer's specification                                                                                                                                                                           |
| Description of measurement methods and procedures to be applied | The cookstoves will be manufactured to match the fixed design specification. This will ensure that ICS delivered to project households is standardized with respect to the dimensions and performance. |
| Frequency of monitoring/recording                               | Once at the time of project stove installation                                                                                                                                                         |
| Value applied                                                   | 7.0 years                                                                                                                                                                                              |
| Monitoring equipment                                            | -                                                                                                                                                                                                      |
| QA/QC procedures to be applied                                  | -                                                                                                                                                                                                      |
| Purpose of data                                                 | -                                                                                                                                                                                                      |
| Calculation method                                              | -                                                                                                                                                                                                      |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comments                                                        | <p>In case of inclusion of other/new models, during project lifetime, corresponding thermal efficiency values shall be used for ER calculations.</p> <p>Also, the PP shall either replace the ICS after the end of its life with an equivalent device to continue claim credits or otherwise shall cease claiming credits after the rated lifespan.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Data / Parameter                                                | $B_{y=1, new, i, survey}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data unit                                                       | tonnes/device/year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Description                                                     | Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i, determined in the first year of the implementation of the project through a sample survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Source of data                                                  | Project fuel measurement campaigns/ survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description of measurement methods and procedures to be applied | Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, $y=1$ ) through fuel measurement campaigns at representative households and/or sample survey.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Frequency of monitoring/recording                               | Determined in the first year of project implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Value applied                                                   | For ex-ante calculation purpose, the value is assumed as<br>0.76 tonnes/device/year for firewood<br>0.33 tonnes/device/year for charcoal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Monitoring equipment                                            | weighing scale, moisture meter etc. in case of fuel measurement campaigns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| QA/QC procedures to be applied                                  | conservative bound value of the results may be used in case desired precision is not achieved,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Purpose of Data                                                 | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Calculation method                                              | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Comments                                                        | <p>Alternatively, sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (instead of fuel measurement campaigns) may only be used if the following conditions are satisfied.</p> <ul style="list-style-type: none"> <li>(i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households;</li> <li>(ii) If multiple devices are used in the project; it is possible from the results of the survey questions to clearly differentiate the quantity of woody biomass being used by each device. In other words, if more than one device, or another device that consumes woody biomass, are in use in project households, then the sample survey needs to distinguish the quantity of woody biomass used by the project device and the other devices that use woody biomass.</li> </ul> |

### 5.3 Monitoring Plan

The monitoring plan is based on the applied methodology VMR0006, Version V1.1.

### Sampling plan

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of Sampling Plan in line with the requirements of the methodology applied and the Standard: Sampling and surveys for CDM project activities and programmes of activities, version 9.0

Sample sizes will be sufficient to ensure that the precision of the sample means/proportions are in accordance with the methodological / standard requirements. In cases where survey results indicate that desired precision is not achieved, the lower bound value of the corresponding confidence interval of the parameter value may be used as an alternative to repeating the survey. Alternatively, the survey may be expanded to reach the required confidence/precision, unless option (b) on page 12 of the VMR0006 is used.

The sampling methodology will be in accordance with the representative sampling methods provided by the sampling guidelines and standards as indicated along this section, with the applied methodology having precedence.

### Sampling Methodology

**Sampling Objective** – The sampling objective for each parameter is to determine, via sampling survey, a statistically significant parameter value for the emission reduction calculations. These parameters are as listed in section 5.2:

| No. | Monitoring Parameters  | Sampling Parameters                               | Parameter type | Monitoring frequency                                                                  |
|-----|------------------------|---------------------------------------------------|----------------|---------------------------------------------------------------------------------------|
| 1   | $N_{y,i,j}$            | Proportion of operational ICS                     | Proportion     | Biennially                                                                            |
| 2   | $B_{y=1,new,i,survey}$ | Quantity of woody biomass used by project devices | Mean           | First year of project implementation and fixed thereafter for entire crediting period |

Field Measurement Objective and data to be collected – As listed in Section 5.2 above.

**Target population and sampling frame** – The target population is the total population served under the project, and the sampling frame consists of aggregated data of end-users of the ICS as recorded in the Project Database. The sampling frame will be kept for 2 years following the crediting period or the last issuance of the CERs of the project activity.

**Sample method** – Sampling will be conducted using stratified random sampling techniques<sup>13</sup> over the sampling frame, and detailed calculations are provided below as per CDM guidelines “Sampling and surveys for CDM project activities and programmes of activities”. The ICSs in the sampling frame shall be stratified as per parameter requirements.

Stratified Random Sampling will be used to select samples from the Project Database for monitoring parameters.

The sample size shall be determined using the following formula:

$$n \geq \frac{z^2 * N * V}{(N - 1) * precision^2 + z^2 * V}$$

<sup>13</sup> Other Sampling approaches may be applied during the course of crediting period, in case needed.

Where,

n = number of ICS to be sampled

N = Total number of ICS in the population

Z = Constant referring to level of confidence (1.96 for 95 %, 1.646 for 90%)

Precision = Required precision (e.g., 10% = 0.1)

Where for proportion-based parameters:

$$V = \frac{SD^2}{p^2}$$

$$SD^2 = \frac{\sum_{i=1}^k g_i * p_i * (1 - p_i)}{N}$$

$$\bar{p} = \frac{\sum_{i=1}^k g_i * p_i}{N}$$

Where,

SD<sup>2</sup> = Weighted Overall Standard deviation

g<sub>i</sub> = weight of strata i in the population

p<sub>i</sub> = expected proportion of strata i in the population

k = total number of strata in the population

And for mean-based parameters:

$$V = \frac{SD^2}{mean^2}$$

$$SD^2 = \frac{\sum g_i \times SD_i^2}{N}$$

$$mean = \frac{\sum g_i \times m_i}{N}$$

Where,

Mean = weighted overall mean

m<sub>i</sub> = Mean of strata i

Optionally, other sampling approaches may be used in accordance with Standard “Sampling and surveys for CDM project activities and programmes of activities” and Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities when sampling techniques or statistical analysis necessitates it.

To ensure a random sample selection, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its Serial ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the CME.

In the case of parameters monitored for the first time the expected variation for that measure in the sample may be based on results from similar studies, pilot studies, or from the project planner’s own knowledge / experience of the data.

The data will be collected using surveys carried out either by external third parties or by project implementer teams. Data collected from the surveys will be compiled into an Excel spreadsheet.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations.

During sampling, there may be non-response from the target population. Over-sampling may be used to avoid non-response; however, sampling may be ceased once required confidence/precision is met.

Alternatively, a simplified approach proposed in option (b) under Section 8.4 of methodology may be used for determining the minimum sample size for usage survey in which case compliance with required confidence/precision is not obligatory. The minimum sample size for project cookstove to conduct the usage survey should be:

1. Project target population < 300: Minimum sample size 30
2. Project target population 300 to 1000: Minimum sample size 10% of group size
3. Project target population > 1000: Minimum sample size 100

Similarly, for fuel measurement campaigns, sample sizes may be alternatively determined as per the available KPT protocols.

Refer section 1.11 above for the management plan and operation framework (e.g., responsibilities and competencies of the personnel that will be carrying out monitoring activities, the procedures for internal auditing and QA/QC, etc.).



## Appendix 1: Non-Binding Mechanisms for Success of the Carbon project

The primary objective of the project is to minimize CO<sub>2</sub> emissions and health hazards associated with traditional inefficient cooking. Through this initiative, the project embodies the core essence of emission reductions by transitioning project beneficiaries from non-renewable biomass fuels to cleaner cooking technologies. This transition contributes to a greener environment while generating valuable carbon credits that ensure project's sustainability. The PP has adopted the following practices to ensure successful implementation of the cookstove carbon project. It is worth noting that these practices / mechanisms are only indicative and may be changed by PP during the course of the project life. The following section, hence, is non-binding on the PP and any change from the following shall not be deemed as a deviation / design change from the registered project activity:

### 1. Atlas Restriction Policy

This policy ensures that each household is entitled to only one project cookstove, registered under a single account. By enforcing this policy, it guarantees accurate distribution and accountability of project cookstoves and eliminates double counting and stove stacking risks.



### 2. Know Your Customer (KYC) Process

The project has a robust "Know Your Customer" process. Through this process, it is verified that the customer purchasing the project cookstove is the intended user. The KYC process also safeguards the authenticity of the project and ensures project cookstoves are exclusively sold to legitimate customers.

### 3. Training for Area Managers and Sales Agents

The PP has prioritized the provision of consistent training for the sales team, including Solar Energy Promoters (SEPs)/Commissioned agents. Through comprehensive training sessions, their understanding of the project's objectives is enhanced. These trainings not only promote transparency and accuracy during the project implementation but also equips the SEPs / commission agents to train customers on effective use of the project cookstoves. By extending the training impact to include customer education, it gets ensured that users are well-informed about proper usage and benefits of the project cookstoves.

### 4. Updated Customer Agreement Contract

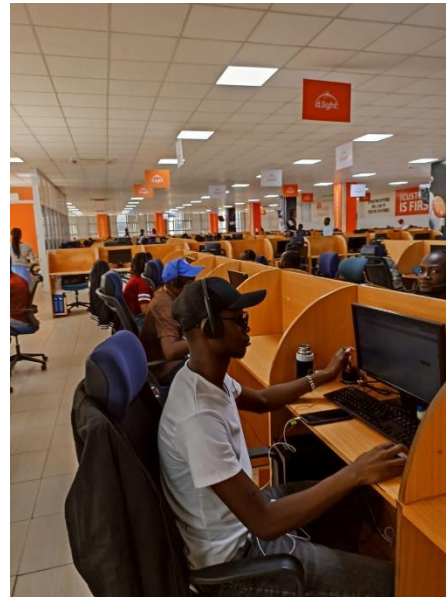
PP has developed a customer agreement contract that includes Carbon Credits ownership transfer to PP and awareness to the end users on the concept. Through this updated contract, customers



acknowledge and grasp the carbon credit aspect of the project. The Rights' transfer signature ensures compliance and alignment with the objectives of the project.

### 5.Weekly, Monthly and Annual Verification.

To ensure the maintenance of high-quality standards, project has a verification process. This process involves conducting periodic checks on the distribution data, guided by a comprehensive questionnaire, supported by the inhouse call center team. The objective is to verify the correctness of distribution transactions and promptly detect and resolve any potential gaps or inconsistencies. Through regular verification of project database, the project's integrity is upheld and associated risks are minimized, safeguarding the overall quality of the project.



### 6.Field Visit

Regular field visits are conducted after distribution to physically inspect project sites, interact with stakeholders, and gather first-hand information on the progress and quality of our cookstove carbon project. Field visits provide valuable insights and enable us to make informed decisions for further improvement and quality assurance.



**7. Customer service**

The PP prioritizes excellent customer service and ensures prompt addressal of any issues or concerns that project beneficiary may have. The established channels for customer support, including dedicated helplines or contact points, enable prompt response to customer inquiries and provide assistance as needed.

**8. Data Collection App**

A data collection app designed to capture essential data, including location details, project statistics, and other relevant information helps regarding the carbon project success. This application simplifies the data collection process, effects improved accuracy and efficiency while recording project-related data.

**9.Frequent Communication/Reminders**

Regular communications with the customers are maintained, including sending SMS messages, to remind them of the benefits they receive from participating in the cookstove carbon project. These messages serve as reminders of the positive impact they are making on the environment and the incentives they earn use of project cookstoves.

## APPENDIX 2: $f_{NRB}$ CALCULATION AS PER CDM TOOL 30, V4.0

### **Applicability Condition as per para 3:**

*This tool may be used by:*

- a) DNAs to submit region- or country-specific default  $f_{NRB}$  values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or
- b) project participants to calculate project or PoA-specific  $f_{NRB}$  values.

**Justification:** A project specific  $f_{NRB}$  value is being calculated.

### **Condition as per para 5:**

*Project participants and DNAs shall identify and clearly delineate the applicable area for which  $f_{NRB}$  is determined. For project- or PoA-specific  $f_{NRB}$  values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project activity and justify the selection of the area in CDM project design documents. For region- or country-specific  $f_{NRB}$  values, DNAs shall specify the applicable area.*

**Justification:** Given the project boundary is geographical boundary of Kenya and the ICSS implemented under the project shall be distributed across Kenya, hence the applicable area being considered is Kenya and the  $f_{NRB}$  value has been calculated for the same.

### **Condition as per para 6:**

*Project participants and DNAs may choose between the following two options to determine the value of  $f_{NRB}$ :*

- a) Use the default value as provided in TOOL33; or
- b) Calculate  $f_{NRB}$  by determining the share of renewable and non-renewable woody biomass in the total quantity of wood biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and requirements in the paragraphs below. The project participants shall compare and analyse the calculated values against the values for  $f_{NRB}$  reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least:

- (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. *Nature Climate Change*, 5(3), pp. 266–272.

**Justification:** PP has opted option (b) to determine the value of  $f_{NRB}$ . The values that have been used in the calculation of  $f_{NRB}$  are directly sourced from latest publicly available, relevant scientific literature. Given the PP has not used any self-collected data, but directly sourced values from scientific published literature, hence a comparison with other sources is not deemed applicable. A detailed description of the data sources and values considered for determining  $f_{NRB}$  is presented in below section.

Further, para 6b) of tool 30 v 4.0 requires comparison of the calculated  $f_{NRB}$  value with other scientific literature wherein the literature **should (and not shall)**, and hence is not deemed mandatory for use) atleast include the following:

- (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. *Nature Climate Change*, 5(3), pp. 266–272.

An assessment of the aforesaid literature (Bailis et.al) yielded the following:

- (i) The report is dated 2015 and does not present a recent scenario in Kenya truly. Further, para 10 of the CDM Tool 30 v.4.0 clearly mentions that “in case of ex ante calculation of

$f_{NRB}$ , the parameter  $f_{NRB}$  shall be estimated using the most recent historical year for which data is available.” Given the PP has calculated  $f_{NRB}$  using more recent reports providing data vintages for 2020-21, hence its comparison with the aforesaid report (Bailis et. al.) being vintaged 2015 is not deemed appropriate.

- (ii) The aforesaid report (Bailis et. al.) under table 1 (page 10-11), cites the source of data used for determining the fuelwood consumption in urban and rural parts of a country. For Kenya specifically, the source used is IEA (reference # 9 and 28), as specified on page 45 of this report, as the following respectively:

- a) International Energy Agency, IEA World Energy Statistics and Balances, <<http://www.oecdilibrary.org/statistics>> (2013)
- b) Ministry of Energy. Study on Kenya's Energy Demand, Supply and Policy Strategy for Households, Small Scale Industries and Service Establishments: Final Report. 158 (KAMFOR Company Limited, Nairobi, 2002)

The aforesaid report (Bailis et. al.) specifies that the IEA Energy Statistics and Balances (2013) values have been adjusted using KAMFOR 2002 report. Thus, these sources necessarily do not reflect the most recent information on fuel consumption in the host country.

- (iii) Similarly, for non-energy use (domestic uses like houses, fencing etc.) consumption, the report (Bailis et. al.) states the following (page 12):

- c) The consumption of construction material for fences, stables, rural houses construction and maintenance ranges between 5 and 20 kg per capita and per year. In this study, a mid-range value of 11 kg per capita year is applied to the rural population. This average is based on references (21, 29 and 33) of the report (Bailis et. al.) as listed on page 45-46. These three references pertain to Rwanda, Mozambique and Sudan respectively and hence are not deemed representative of Kenya.

- (iv) Also, the inclusion of industrial roundwood consumption (specially for non-energy purposes) in determining the  $f_{NRB}$  is not stated in the report (Bailis et. al.) while para 17 of the CDM tool 30 v 4.0 clearly requires its inclusion.

Also, the PP has assessed the established value with other VERRA VCS registered projects. For a comparative analysis, the projects that have their boundary comparable to the “Project” are being considered as follows:

| Project ID  | Applied Methodology    | $f_{NRB}$ Value established | Comparison with the project activity                                                                                                                    |
|-------------|------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>3206</u> | VMR0006 version 1.1    | 94%                         | <b>This Project with the <math>f_{NRB}</math> value of 78.75% is more conservative compared to all other projects already approved under VERRA VCS.</b> |
| <u>2349</u> | VMR0006 version 1.1    | 94%                         |                                                                                                                                                         |
| <u>1918</u> | AMS-II.G., version 9.0 | 91.6%                       |                                                                                                                                                         |
| <u>1082</u> | AMS-II.G., version 3.0 | 96%                         |                                                                                                                                                         |
| <u>2836</u> | AMS-II.G., version 5.0 | 97%                         |                                                                                                                                                         |
| <u>2837</u> | AMS-II.G., version 8.0 | 97%                         |                                                                                                                                                         |
| <u>3197</u> | VMR0006 version 1.1    | 89%                         |                                                                                                                                                         |

**Thus, the  $f_{NRB}$  calculation conducted by PP using relevant scientific literature is in compliance with para 6(b) of the CDM Tool 30, v 4.0.**

**Condition as per para 7:**

*The fraction of woody biomass that can be established as non-renewable is:*

$$f_{NRB} = NRB / (NRB + RB)$$

*Equation (1)*

| <b>Justification:</b> The $f_{NRB}$ calculated follows Equation (1) as above. Refer below for details: |                                                         |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                       |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Data Symbol</b>                                                                                     | <b>Parameter Description</b>                            | <b>Units</b> | <b>Value</b> | <b>Reference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Source</b>                                                                                         |
| <b>F</b> Forest, i                                                                                     | Extent of forest in the relevant period (ha)            | ha           | 3,611,090    | FAO Global Forest Resources Assessment 2020, Kenya, (herein after referred as 'FAO GFRA 2020 Kenya') Section 1a), page 13, value considered for year 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="https://www.fao.org/3/cb0019en/cb0019en.pdf">https://www.fao.org/3/cb0019en/cb0019en.pdf</a> |
| <b>F</b> Other wooded land, i                                                                          | Extent of other wooded land in the relevant period (ha) | ha           | 32,271,450   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                       |
| <b>F</b> Other land, i                                                                                 | Extent of other land in the relevant period (ha)        | ha           | 21,031,460   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                       |
| <b>P</b> Forest, i                                                                                     | Forest area within Protected Areas                      | ha           | 1,178,470    | FAO GFRA 2020 Kenya Section 3b), page 46, values considered for year 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                       |
|                                                                                                        | Forest area within long term Management Plan            | ha           | 152,790      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                       |
| <b>P</b> Other wooded land, i                                                                          | Protected Areas in Other Wooded land                    | ha           | 0            | <p>The FAO GFRA 2020 Kenya does not provide information on the extent of protected areas under "other wooded land" category. <b>As a conservative measure the value for this category has been considered as 0</b> (given this parameter contributes negatively to the quantification of renewable biomass).</p> <p>It is worth noting that the total protected area in Kenya as per Protected Planet is <a href="https://www.protectedplanet.net/country/KEN">https://www.protectedplanet.net/country/KEN</a> 7,153,400 ha which is not being discounted from the "Other Wooded Land" conservatively.</p> | --                                                                                                    |
| <b>P</b> Other land, i                                                                                 | Protected Areas in Other land                           | ha           | 0            | <p>The FAO GFRA 2020 Kenya, does not provide information on the extent of protected areas under "other land" category. <b>As a conservative measure the value for this category has been considered as 0</b> (given this parameter contributes negatively to the quantification of renewable biomass).</p> <p>It is worth noting that the total protected area in Kenya as per Protected Planet is <a href="https://www.protectedplanet.net/country/KEN">https://www.protectedplanet.net/country/KEN</a> 7,153,400 ha which is not being discounted</p>                                                    | --                                                                                                    |



|                                 |                                                                                                        |                         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                                                                                        |                         |      | from the "Other Land" conservatively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                           |
| <b>MAI forest, i</b>            | Mean Annual Increment of woody biomass growth per hectare in forest areas in the relevant period       | m <sup>3</sup> /ha/year | 0.79 | <p>The FAO GFRA 2020 Kenya defines Forests (public natural or private natural) as a group of trees whose crowns are largely contiguous and include the ecosystem that makes it up to a tree canopy cover of over 15%, minimum area of 0.5 ha and minimum height of 2m or with potential to reach 2m (refer Global Forest Resources Assessment 2020, Kenya, Section 1b), page 18)</p> <p>In Kenya, the closed forests, woodlands and bushlands are defined as ecosystem that makes it up to a canopy cover of over 40%, and a minimum height of &gt; 3m (refer <a href="https://portals.iucn.org/library/efiles/documents/2000-019-05.pdf">https://portals.iucn.org/library/efiles/documents/2000-019-05.pdf</a>, Section 2, page 2), thus, all falling in the definition of Forests as per FAO GFRA 2020, Kenya.</p> <p>Thus, an average value for closed forest (1.3 m<sup>3</sup>/ha/year), woodlands (0.64 m<sup>3</sup>/ha/year) and bushlands (0.44 m<sup>3</sup>/ha/year) has been considered for forests.</p> | <a href="https://energyprojectedia.info/wiki/Biomass_Energy_Resources_in_Kenya">https://energyprojectedia.info/wiki/Biomass_Energy_Resources_in_Kenya</a> |
| <b>MAI other wooded land, i</b> | Mean Annual Increment of woody biomass growth per hectare in other wooded lands in the relevant period | m <sup>3</sup> /ha/year | 0.25 | <p>The FAO GFRA 2020 Kenya defines "Wooded land" as land not classified as "Forest" spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of 5-10% or trees able to reach these thresholds; or with a combined cover of shrubs bushes and trees above 10% (refer Global Forest Resources Assessment 2020, Kenya, Section 1b), page 18)</p> <p>In Kenya, the wooded grasslands are defined as land covered with grasses and other herbs, with woody plants covering between 10% - 40% of the ground (refer <a href="https://portals.iucn.org/library/ef">https://portals.iucn.org/library/ef</a></p>                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |

|                                     |                                                                                                 |                         |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                     |                                                                                                 |                         |                                                                | <a href="#">iles/documents/2000-019-05.pdf</a> , Section 2, page 2).<br><br>Thus, the value for wooded grasslands (0.25 m <sup>3</sup> /ha/year) has been considered for other wooded land.                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                           |
| MAI <sub>other land, i</sub>        | Mean Annual Increment of woody biomass growth per hectare in other lands in the relevant period | m <sup>3</sup> /ha/year | 0.08                                                           | The FAO GFRA 2020 Kenya calculates “Other land” as Total Land net of “Forest” and “Other wooded land” (refer Global Forest Resources Assessment 2020, Kenya, Section 1a), page 13)<br><br>A MAI of 0.08 m <sup>3</sup> /ha/year (grasslands) has been assumed for other land. Consideration of 100% of “Other Land” for calculation of renewable biomass is deemed highly conservative given only a certain proportion of “other land” provide for supply of woody biomass.<br><br>This is additionally conservative as the protected areas within “other land” is already considered 0 above. |                                                                                                                           |
| RB                                  | Annual Renewable Biomass                                                                        | m <sup>3</sup> /year    | 11,559,044                                                     | $RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Calculated, as per equation (4), para 19 of applied tool                                                                  |
| H                                   | Total consumption of woody biomass (H)                                                          | m <sup>3</sup> /year    | 54,844,000 (year 2020)<br>54,406,000 (year 2021)<br>54,406,000 | Sourced from UN data, the total woody biomass consumption in Kenya (total Fuelwood energy supply) has been considered<br><br>The value of year 2021 is being used (being lower) as a conservative measure.                                                                                                                                                                                                                                                                                                                                                                                     | <a href="http://data.un.org/Data.aspx?d=EDATA&amp;f=cmlID%3aFW">http://data.un.org/Data.aspx?d=EDATA&amp;f=cmlID%3aFW</a> |
| NRB = H - RB                        | Non-renewable Biomass                                                                           | m <sup>3</sup> /year    | 42,846,956                                                     | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Calculated                                                                                                                |
| f <sub>NRB</sub> = NRB / (NRB + RB) | Fraction of non-renewable biomass                                                               | Percentage              | 78.75%                                                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Calculated                                                                                                                |

**Condition as per para 8:**

The relevant period should be one year, unless the methodology referring to this tool specifies otherwise.

**Justification:** The f<sub>NRB</sub> calculated is based on data vintage 2020/2021 and represents one year.

**Condition as per para 9:**

The value of f<sub>NRB</sub> for the applicable area shall be calculated using either of the two following options:



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) <i>Ex ante</i>: the <math>f_{NRB}</math> value is determined once at the validation stage, thus no monitoring and recalculation of the <math>f_{NRB}</math> value during the crediting period is required;</p> <p>b) <i>Ex post</i>: the <math>f_{NRB,y}</math> value is determined for the year <math>y</math> in the crediting period, requiring the <math>f_{NRB}</math> value to be updated annually, following a consistent calculation procedure throughout the crediting period.</p>                                                                                                                                                                                                                                                                                                                  |
| <p><b>Justification:</b> The <math>f_{NRB}</math> calculated is being fixed ex-ante.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Condition as per para 10:</b><br/> <i>In the case of ex ante calculation of <math>f_{NRB}</math>, the parameter <math>f_{NRB}</math> shall be estimated using the most recent historical year for which data is available. In the case of ex-post calculation of <math>f_{NRB}</math>, the parameter <math>f_{NRB}</math> shall be estimated for the applicable year <math>y</math> of the crediting period. Where available, the same vintage of data should be used for all parameters applied in this tool to calculate <math>f_{NRB}</math>. Where data for one single vintage is not available for all parameters, different vintages may be used for parameters, as long as it can be justified (e.g. the use of different vintages leads to a conservative estimate of <math>f_{NRB}</math>).</i></p> |
| <p><b>Justification:</b> The <math>f_{NRB}</math> is being calculated ex-ante. As per above, the data vintage used is most recent available, pertaining to 2020-21.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Condition as per para 11:</b><br/> <i>The quantity of non-renewable biomass consumed in the applicable area (NRB) shall be determined as the difference between the total consumption of woody biomass in the applicable area (H) and the quantity of renewable biomass that can be sustainably harvested in the applicable area (RB):</i><br/> <math display="block">NRB = H - RB \quad \text{Equation (2)}</math> </p>                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Justification:</b> The NRB has been calculated using above equation. Refer information under the “condition as per para 7” above.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Condition as per para 12:</b><br/> <i>Cubic meters (<math>m^3</math>) instead of tonnes (t) of wood may be used for estimation of <math>f_{NRB}</math> values, as long as the same unit and the same conversion factor (e.g. wood density, moisture content) are consistently used for estimation of both the total consumption of woody biomass (H) and the renewable biomass available (RB).</i></p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>Justification:</b> The total consumption of woody biomass (H) and the renewable biomass available (RB) both are using cubic meters as the unit. The total woody biomass consumption is sourced from UN Data (<a href="http://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3aFW">http://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3aFW</a>) and is presented in cubic meters. For renewable biomass estimations, the Mean Annual Increment (MAI) values (<math>m^3/ha/year</math>) under different land use categories are multiplied with the respective land areas (ha). Hence, both the values are having same units and can be used to establish the <math>f_{NRB}</math> values.</p>                                                                                                                          |
| <p><b>Condition as per para 13:</b><br/> <i>If the <math>f_{NRB}</math> value is estimated at the national level, as a cross check, project proponent shall compare the value of estimated NRB with the product of:</i> <ol style="list-style-type: none"> <li><i>total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and</i></li> <li><i>most recent available observed annual rate of deforestation (ha/yr).</i></li> </ol> <i>If the estimated NRB value is more than 10% above the value calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB.</i> </p>                                                                                                                     |
| <p><b>Justification:</b><br/> The expansion of forest cover, as presented in FAO GFRA 2020 Kenya, is in contradiction with the results established above and deforestation reported in other relevant published literature. FAO GFRA 2020 Kenya reported an average net gain in forest area of 17,740 ha with increase in naturally regenerating forest between 2015 and 2020 (refer FAO GFRA 2020 Kenya, Section 1d, page 27). The reported forest expansion rate does not account deforestation estimates in the FAO GFRA 2020 Kenya as shown below:</p>                                                                                                                                                                                                                                                         |

| FRA 2020 report, Kenya        |                     |           |           |           |
|-------------------------------|---------------------|-----------|-----------|-----------|
| FRA categories                | Area (1000 ha/year) |           |           |           |
|                               | 1990-2000           | 2000-2010 | 2010-2015 | 2015-2020 |
| Forest expansion (a)          |                     |           |           |           |
| ...of which afforestation     |                     |           |           |           |
| ...of which natural expansion |                     |           |           |           |
| Deforestation (b)             |                     |           |           |           |
| Forest area net change (a-b)  | 10.27               | -34.49    | -18.79    | 17.74     |

On the contrary, the National Report titled, “The National Forest Reference Level for REDD+ Implementation”, August 2020, by Ministry of Environment and Forestry, Republic of Kenya reports 338,863 ha/year as the annual deforestation rate in the country (Refer [https://redd.unfccc.int/media/kenya\\_national\\_frl\\_report-august\\_2020.pdf](https://redd.unfccc.int/media/kenya_national_frl_report-august_2020.pdf), “Table-8” on page 29).

It appears, therefore that the forest area net change reported in FAO GFRA 2020 Kenya, does not present a correct value of deforestation rate compared to actual biomass consumption in the country. Further, please refer below:

| Parameter Description                                 | Value                  | Unit           | Reference                                                                                                                                                                                                                                                                                                                                                                                                                                             | Source                                                                                                                                                        |
|-------------------------------------------------------|------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average net forest lost per year                      | 338,863                | ha/year        | Report “The National Forest Reference Level for REDD+ Implementation”, August 2020, by Ministry of Environment and Forestry, Republic of Kenya, page 29, table 8, shows that the long term annual average rate of deforestation in Kenya corresponds to 338,863 ha/yr. <b>Also, even if the lowest rate of deforestation (265,687 ha/yr) as per table 8 below is considered, the compliance with para 13 of CDM tool 30, v4.0 is not compromised.</b> | <a href="https://redd.unfccc.int/media/kenya_national_frl_report-august_2020.pdf">https://redd.unfccc.int/media/kenya_national_frl_report-august_2020.pdf</a> |
| Above Ground Biomass (AGB) as per FAO GFRA 2020 Kenya | 200.38                 | tonnes/ha      | FAO GFRA 2020 Kenya Section 2c), page 40, values considered for year 2020                                                                                                                                                                                                                                                                                                                                                                             | <a href="https://www.fao.org/3/cb0019en/cb0019en.pdf">https://www.fao.org/3/cb0019en/cb0019en.pdf</a>                                                         |
| AGB lost per year                                     | 67,901,368             | tonnes         | Calculated                                                                                                                                                                                                                                                                                                                                                                                                                                            | --                                                                                                                                                            |
| AGB lost per year                                     | 116,835,596            | m <sup>3</sup> | Applying a density factor of 0.581, calculated as average of 282 tree species in Tropical Africa                                                                                                                                                                                                                                                                                                                                                      | Calculated using <a href="http://www.fao.org/3/W4095E/w4095e0c.htm">http://www.fao.org/3/W4095E/w4095e0c.htm</a>                                              |
| Estimated NRB in the f <sub>NRB</sub> calculations    | 42,846,956             | m <sup>3</sup> | The Estimated NRB is less than the AGB lost per year (based on deforestation rate and AGB/Ha), substantiating that the <b>value of f<sub>NRB</sub> established by the project is highly conservative.</b>                                                                                                                                                                                                                                             | --                                                                                                                                                            |
| Check for para 13 compliance                          | Para 13 compliance met | --             | Calculated                                                                                                                                                                                                                                                                                                                                                                                                                                            | --                                                                                                                                                            |

**Table 8: Annual transitions (No of ha); Deforestation and Afforestation among forest strata**

| Forest strata                 | Area (ha/yr) of Deforestation |           |           |           |         |
|-------------------------------|-------------------------------|-----------|-----------|-----------|---------|
|                               | 2002-2006                     | 2006-2010 | 2010-2014 | 2014-2018 | Average |
| Montane & Western Rain Forest | 104,874                       | 72,059    | 72,648    | 76,322    | 81,476  |
| Coastal & Mangrove Forest     | 50,388                        | 27,463    | 52,359    | 56,664    | 46,719  |
| Dryland Forest                | 213,787                       | 166,164   | 258,443   | 204,279   | 210,668 |
| Total                         | 369,049                       | 265,687   | 383,450   | 337,265   | 338,863 |